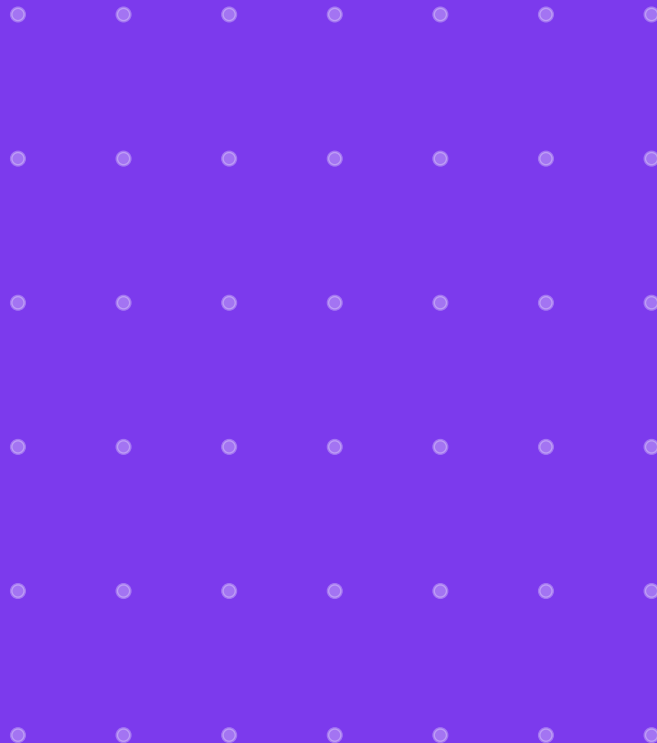


Build & Secure GenAI Applications with AWS Bedrock: Bootcamp

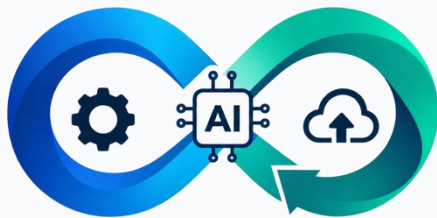
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Amazon Bedrock Course



Build production-ready
Generative AI systems on AWS



- Foundational prompting strategies
- Multimodal API integration
- Visual workflow orchestration with Bedrock Flows
- Enterprise agent deployment using AgentCore
- Retrieval-Augmented Generation (RAG)



Madhuri Jha

DevOps Specialist

Thinknyx®

 **ThinkRook**
By Thinknyx Technologies LLP



Yogesh Raheja

Founder & Senior
Solutions Architect

How Will This Course Work?

Foundations of Amazon Bedrock

- Set Up AWS CLI & Console Prerequisites
- Explore Amazon Bedrock Foundation Models
- Understand Model Configurations & Capabilities
- Test Models in the Bedrock Playground

Programmatic Integration with Python

- Getting Started with AWS SDK (Boto3)
- Mastering InvokeModel & Converse APIs
- Exploring Multimodal Image Analysis
- Building an Interactive AI Chatbot

Prompting Techniques & Prompt Management

- Zero-Shot & Few-Shot Prompting
- Chain of Thought(CoT) & ReAct
- Role-Based Prompting Techniques
- Structured Output Prompting

Embeddings & Knowledge Bases

- Understanding Tokens & Embeddings
- Exploring RAG Architecture
- Building Bedrock Knowledge Bases
- Automated Bedrock Knowledge Bases

How Will This Course Work?

Visual Orchestration with Flows

- Introduction to Bedrock Flow Builder
- Working with Prompt & Knowledge Base Nodes
- Adding Conditions, Loops & Iterators
- Integrating AWS Lambda Functions

Advanced Bedrock & AgentCore

- Building Custom AI Agents with AgentCore
- Integrating Knowledge Gateways & Python CLIs
- Enforcing Bedrock Guardrails & Customization
- Optimizing Cost & Latency Performance

Capstone Project (HR Document Q&A System)

- Building an Enterprise RAG Pipeline
- Vector Search with OpenSearch Serverless
- Deploying with AgentCore CLI
- Testing, Validation & Observability



Equipped to architect, build, and deploy production-grade Generative AI applications using **Amazon Bedrock** and Python

Build & Secure GenAI Applications with AWS Bedrock: Bootcamp

Yogesh Raheja · Thinknyx Technologies LLP



ThinkRook
By Thinknyx Technologies LLP

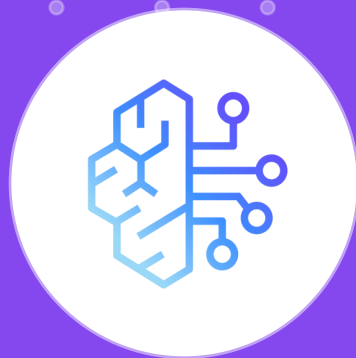


Amazon Bedrock Course

SECTION INTRODUCTION

Introduction to Amazon Bedrock

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 1 — Introduction to Amazon Bedrock

Introduction to Amazon Bedrock

Everything you need before you build your first agent.

1.0

Section Introduction

1.1

What is Amazon Bedrock

1.2

Key Features & Functionalities

1.3

Demonstration - Pre-requisites setup (AWS console & CLI)

1.4

Demonstration - Bedrock Console Navigation

1.5

Use cases & Industry Applications

What is Amazon Bedrock

Section 1 › Introduction to Amazon Bedrock

What is Amazon Bedrock

Building these systems traditionally is not simple



AI-Powered Applications

Chatbots



Document
assistants



Content
generators



Recommendation
systems



AI agents



What is Amazon Bedrock

GPU Infrastructure

**Machine
learning expertise**

Large AI models

What is Amazon Bedrock

GPU Infrastructure

Machine
learning expertise

Large AI models

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GPU Infrastructure

Machine
learning expertise

Large AI models

What is Amazon Bedrock

Scaling mechanisms

Machine
learning expertise

Large AI models

What is Amazon Bedrock

GPU Infrastructure

APIs

Large AI models

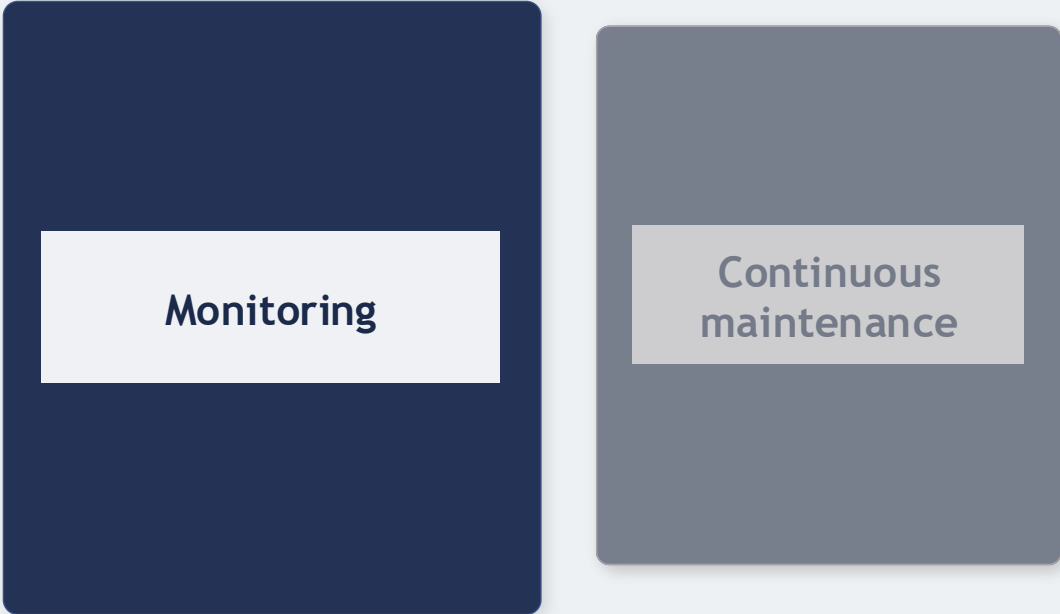
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Security

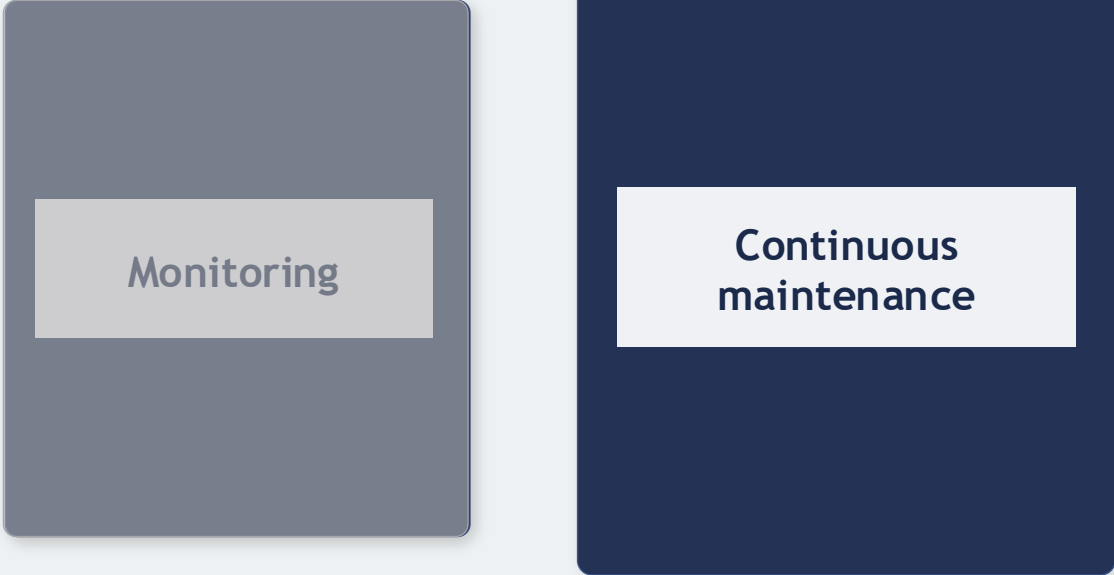
What is Amazon Bedrock



Monitoring

**Continuous
maintenance**

What is Amazon Bedrock



Monitoring

Continuous
maintenance

What is Amazon Bedrock



Amazon Bedrock is a fully managed AWS service that gives you access to powerful foundation models

What is Amazon Bedrock

Core building blocks
of Generative AI

- ✓ Downloading models
- ✓ Configuring GPUs
- ✓ Managing servers
- ✓ Building infrastructure



**AI capabilities in your
applications**

What is Amazon Bedrock



Unified gateway to
Generative AI innovation

AWS service

AI

Chatbots

stability.ai

Stability AI



Meta



Amazon Titan
models

What is Amazon Bedrock

Focus on building your application

AWS handles the infrastructure for you

Don't provision
GPUs

Don't configure
model pipelines

Don't maintain
servers

Don't worry about
scaling

What is Amazon Bedrock



Customer support chatbot

Without Bedrock



- ✓ Expensive
- ✓ Complex
- ✓ Time-consuming



GPU-based
servers



Large language
model



Configure
APIs



Manage
dependencies



Optimize
performance

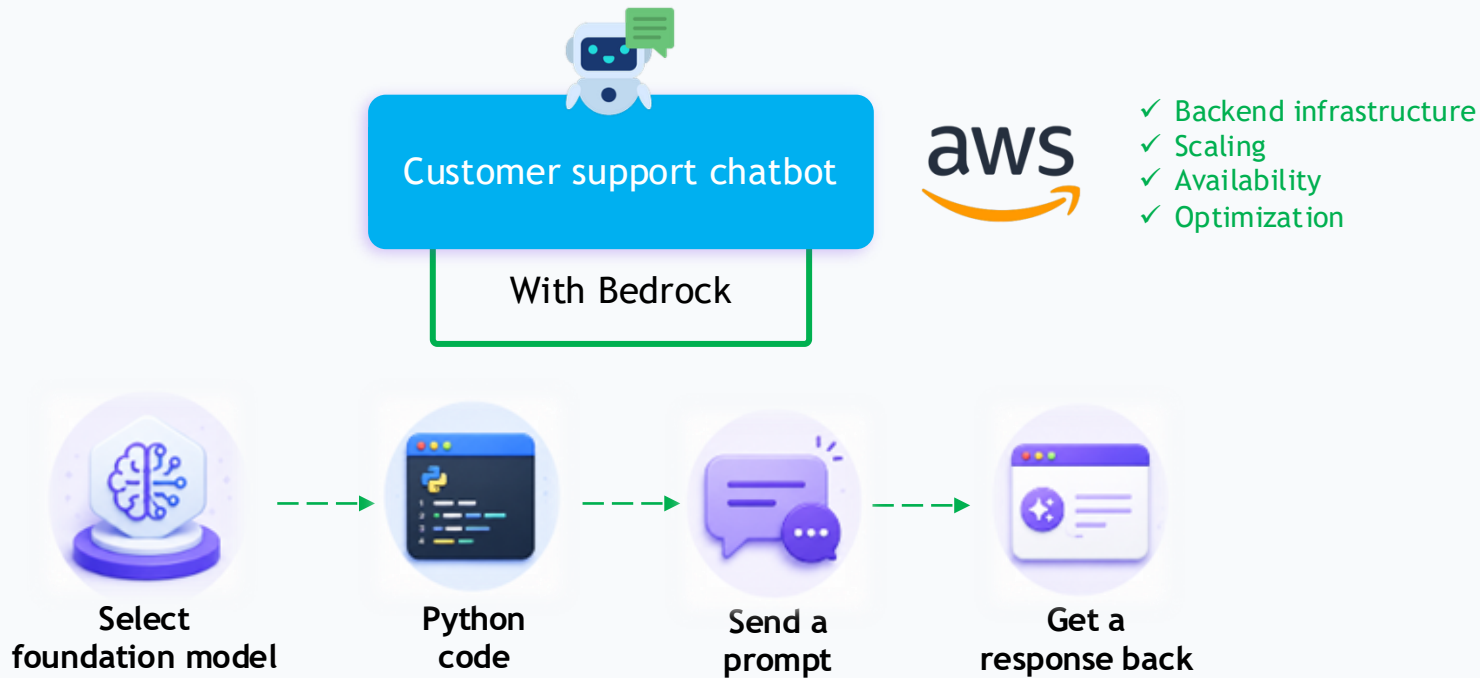


Secure
infrastructure



Continuously
maintain everything

What is Amazon Bedrock



What is Amazon Bedrock



Advanced
reasoning tasks



Open-source
flexibility



Cost optimization

- ✓ Learning different APIs
- ✓ Creating accounts on multiple platforms
- ✓ Handling separate billing systems
- ✓ Rewriting parts of your application

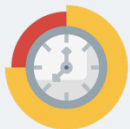
What is Amazon Bedrock



Models are available through one AWS service

Makes experimentation much faster and easier for developers and organizations

What is Amazon Bedrock



Without Managed Platform



Buy hardware

Configure wiring

Install software

Troubleshoot issues

Maintain everything

What is Amazon Bedrock

With Managed Platform

Everything is already integrated
and ready to use



What is Amazon Bedrock



Works the same way for
Generative AI



Handles the infrastructure
and operational complexity

What is Amazon Bedrock



Works the same way for
Generative AI



Focus on building
applications and business
solutions



Handles the infrastructure
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What is Amazon Bedrock



Allows developers and organizations to build AI-powered applications faster

Lower cost

Reduced complexity

Less operational overhead

Key Features & Functionalities

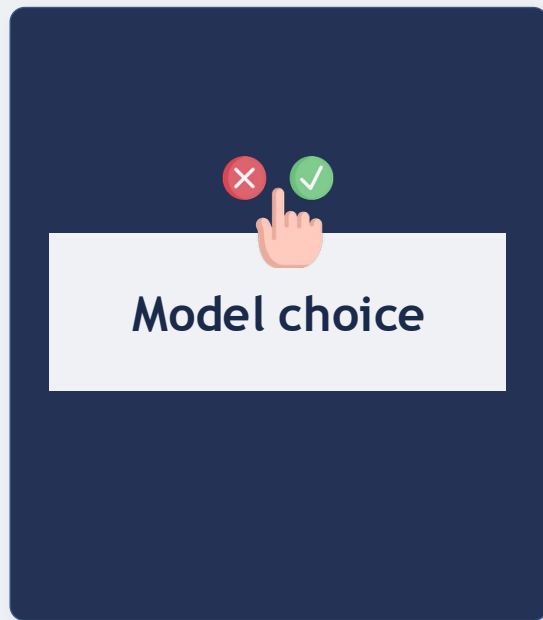
Section 1 › Introduction to Amazon Bedrock

Key Features



**What exactly can we do with
Amazon Bedrock?**

Key Features



Key Features



Model choice

➤ **Generative AI** :- today is that different models are good at different tasks

Better at reasoning

Optimized for speed

Cost-effective

Content generation

Coding assistance

Key Features

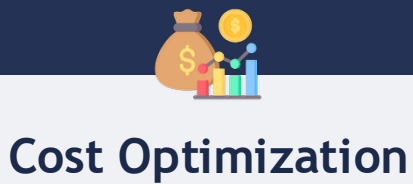


Model choice

- **Amazon Bedrock** :- gives you access to multiple foundation models from leading AI providers through one platform



Key Features



Key Features



Cost Optimization

➤ Bedrock offers several intelligent features that let you optimize costs without compromising performance

Model Distillation

Deploy lighter, more efficient versions of larger models

Intelligent Prompt Routing

Automatically directs requests to the most appropriate and cost-effective model based on task complexity

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Reduces redundant API calls by reusing cached prompt contexts

Key Features



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- **Choose Based on Your Workload** — Pick the mode that fits your performance and cost needs

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Key Features



Retrieval-Augmented Generation / RAG

- AI models often need access to company-specific information such as



Internal documents



PDFs



Policies



Databases



Knowledge portals

- Bedrock allows you to securely connect your foundation models to enterprise data sources like

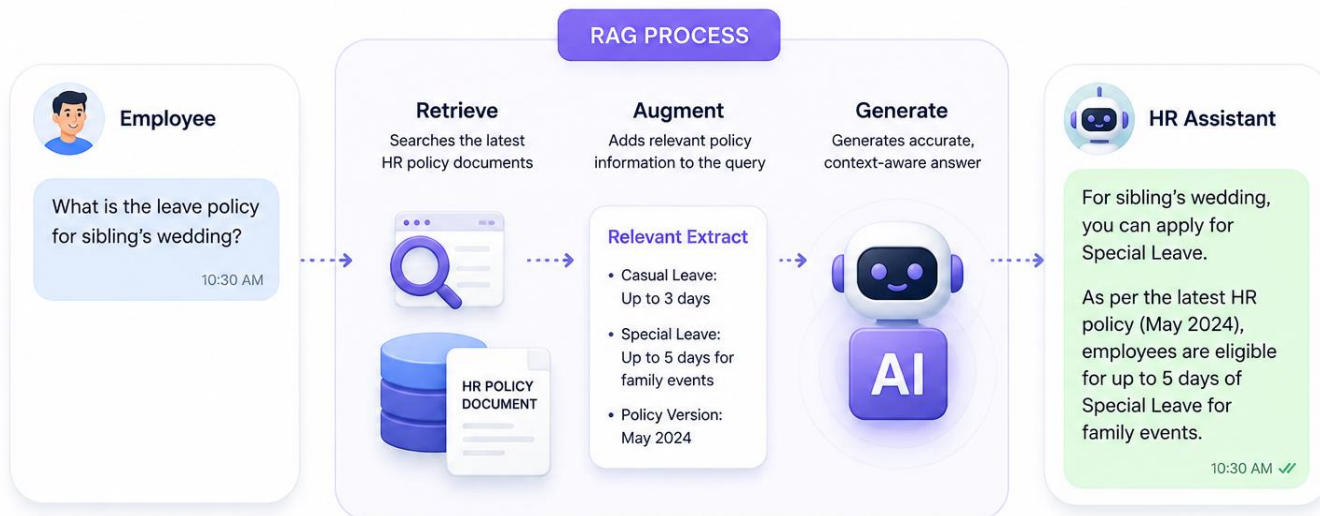
Amazon S3

SharePoint

Salesforce

Knowledge Bases

HR Support Assistant with RAG



Key Features



Key Features



**Amazon
Bedrock Agent Core**

- Comprehensive framework designed to deploy and operate AI agents securely and at scale
- Supports various use cases

Personal assistants

Enterprise automation

Customer service

Data analytics

DevOps

Key Features



Customization capabilities

General-purpose models may not understand your business-specific terminology and processes

Fine-tune models with your data for more business-specific responses

Import custom or open-source models with centralized AWS management and version control

Key Features



Guardrails

Reduce Hallucinations & Harmful Content – Filter unsafe, restricted, or inaccurate responses

Custom Safety Rules – Block sensitive information, keywords, and inappropriate topics

Safer AI Responses – Ensure outputs align with your organization's policies and requirements

Key Features



Data automation

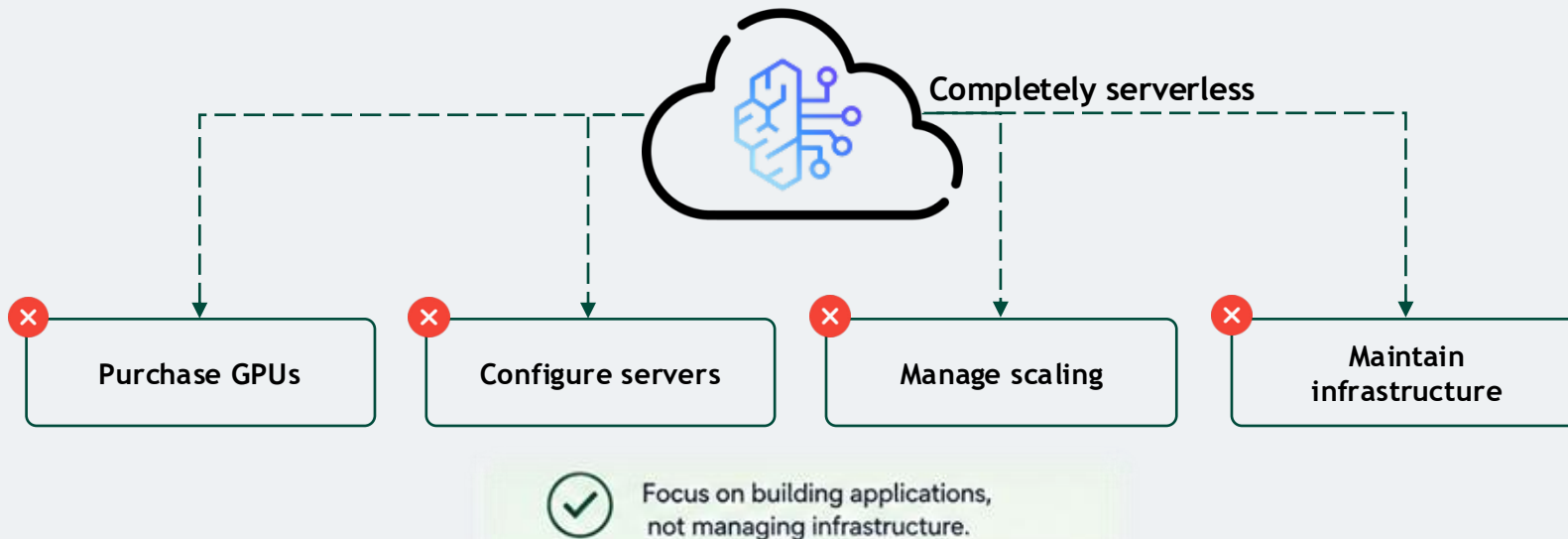
Process Unstructured Data — Analyze documents, images, audio, videos, invoices, and reports

Extract Business Insights — Summarize, extract, and convert content into structured information

Multi-Agent Orchestration — Coordinate multiple AI agents to handle complex workflows

Foundational Benefits of Amazon Bedrock

Zero infrastructure Management



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Proprietary business data
remains private and secure



AWS does not use your data to
train public foundation models

Supports enterprise compliance
(ISO, SOC, HIPAA and GDPR)

Foundational Benefits of Amazon Bedrock

Bedrock integrates seamlessly with the broader AWS ecosystem



Amazon S3
for storage



IAM for
access control



Build production-ready AI
applications



AWS Lambda for
serverless workflows



Amazon CloudWatch
for monitoring

Pre-requisites Setup (AWS console & CLI)

Section 1 › Introduction to Amazon Bedrock

Pre-requisites Setup (AWS Console & CLI)

1



Create an
IAM user

2



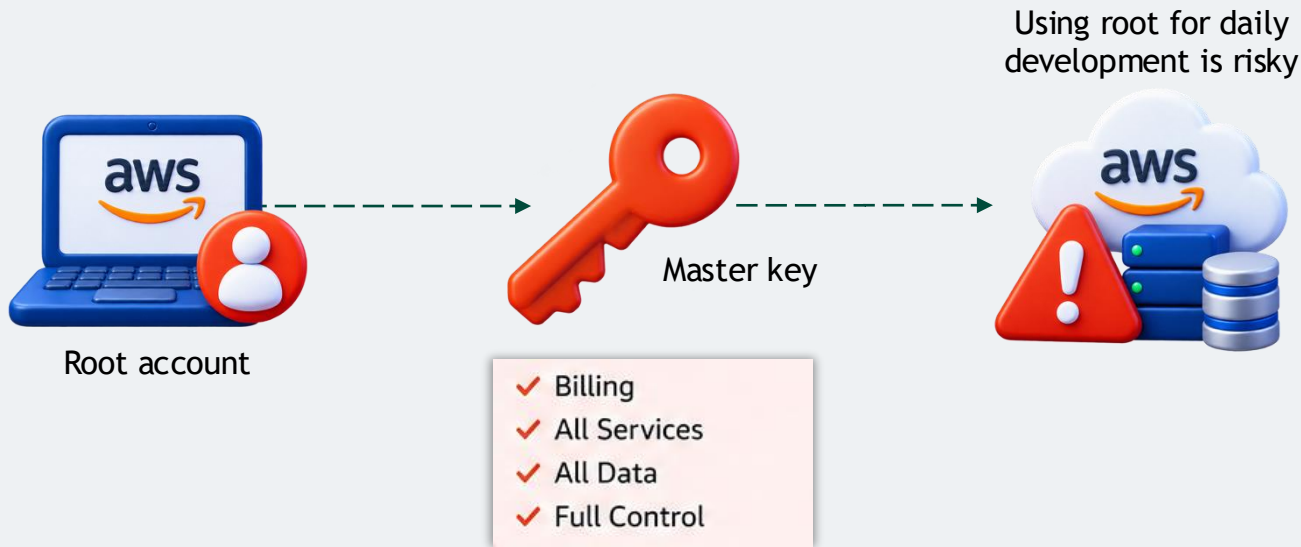
Generate
access keys

3

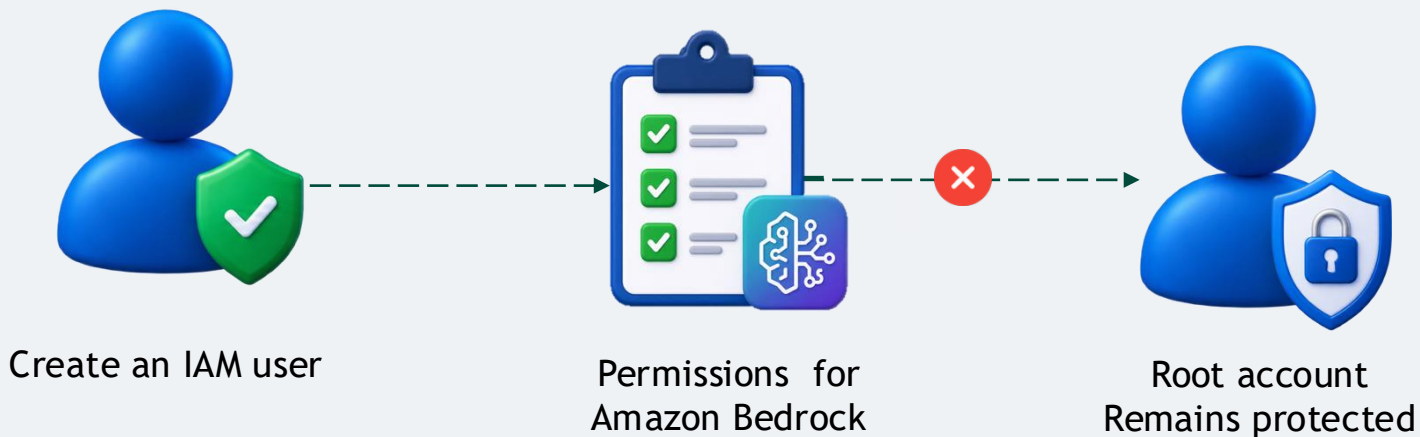


Install & configure
the AWS CLI

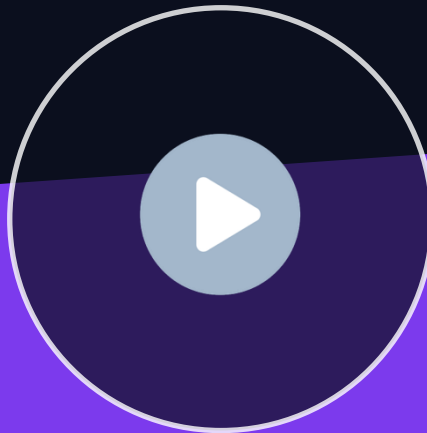
Understanding Why IAM Matters



Understanding Why IAM Matters



L I V E D E M O



Bedrock Console Navigation

Use cases & Industry Applications

Section 1 › Introduction to Amazon Bedrock

Use cases & Industry Applications



Amazon Bedrock



Capabilities it provides

Use cases & Industry Applications



**Where can we actually use
Amazon Bedrock?**

BUSINESS NEED	WHAT AI DOES	EXAMPLE
1 Customer Service	Answer questions	“Package damaged?”
2 Healthcare	Summarize information	Doctor-patient conversation
3 Financial Services	Analyse reports	Structured key information
4 Human Resources	Answer policy questions	“Paternity leave?”
5 Retail & E-Commerce	Create product content	Product descriptions
6 Legal & Compliance	Search documents	“Termination clauses?”
7 Software Development	Help with code	Explain functions / tests
8 Education & Training	Teach & practice	Simpler explanations
9 Media & Content	Create drafts	Blogs • emails • social

Use cases & Industry Applications

1

Document Intelligence

Understand information
and answer questions from
documents

2

Automation

Automate repetitive
tasks and workflows

3

Content Generation

Generate text, code,
summaries, or other content
at scale

SECTION INTRODUCTION

Getting started with Amazon Bedrock

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 2 – Getting started with Amazon Bedrock

Getting started with Amazon Bedrock

Step into the Playground and bring your prompts to life.

2.0

Section Introduction

2.1

Foundation models available on Bedrock

2.2

Use the Bedrock Playground to test models interactively

2.3

Explore important model parameters and configuration settings

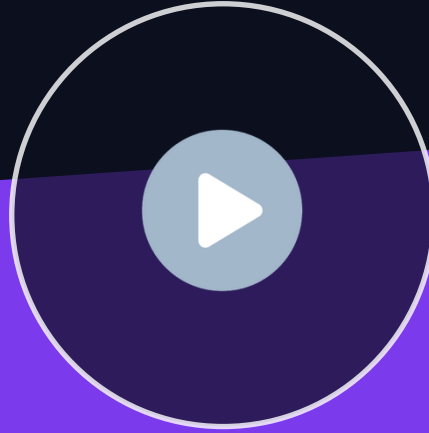
2.4

Understand inference profiles and batch inference concepts

2.5

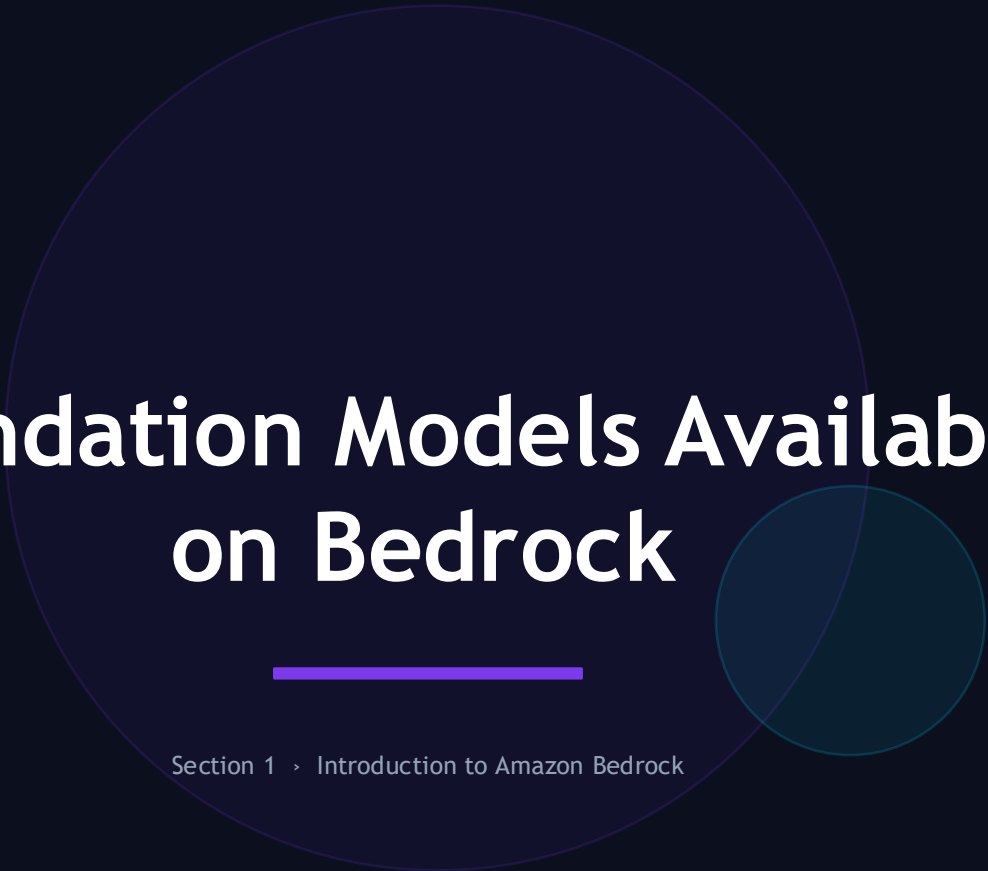
Practical demonstration of inference profiles

L I V E D E M O



Foundation Models Available on Bedrock

Foundation Models Available on Bedrock



Section 1 › Introduction to Amazon Bedrock

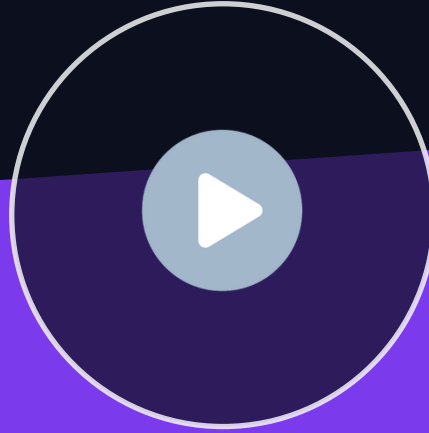
Model Comparison Matrix

Model Type	Best For	Strengths	Considerations
○ Nova 2 Lite	Chatbots, high-volume applications	Fast & cost-effective	Less advanced reasoning
○ Nova Pro	Business applications	Balanced performance & cost	Higher cost than Lite
○ Claude Models	Reasoning, summarization, analysis	Strong reasoning and accuracy	May have higher inference costs
○ Llama Models	General-purpose AI applications	Flexible and widely adopted	Performance varies by model size
○ Embedding Models	Search, semantic search, & RAG applications	Generates vector embeddings	Not used for text generation

Model Selection Guide

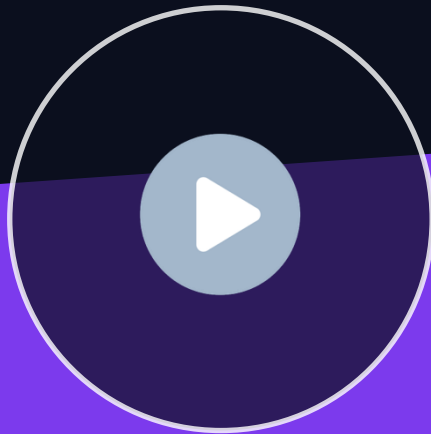
- ✓ **Chatbots** — Prioritize speed and cost efficiency
- ✓ **Document & Research Apps** — Prioritize reasoning and context
- ✓ **Image + Text Apps** — Choose multimodal models
- ✓ **Search & Recommendations** — Use embedding models

L I V E D E M O



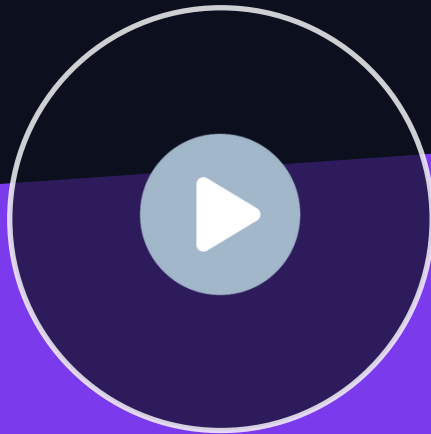
Foundation Models Available on Bedrock

L I V E D E M O



Test with Playground

L I V E D E M O



Exploring Model Parameters and configuration

Inference profiles & Batch inferences

Section 1 › Introduction to Amazon Bedrock

Inference profiles & Batch inferences



What if you're building a production application that serves users across different regions?

What if you need to process thousands of prompts instead of just one?

Inference profiles & Batch inferences

**Inference
Profiles**

**Batch
Inference**

What are Inference Profiles?



Inference Profiles



Instead of calling a model directly, your application can use an inference profile

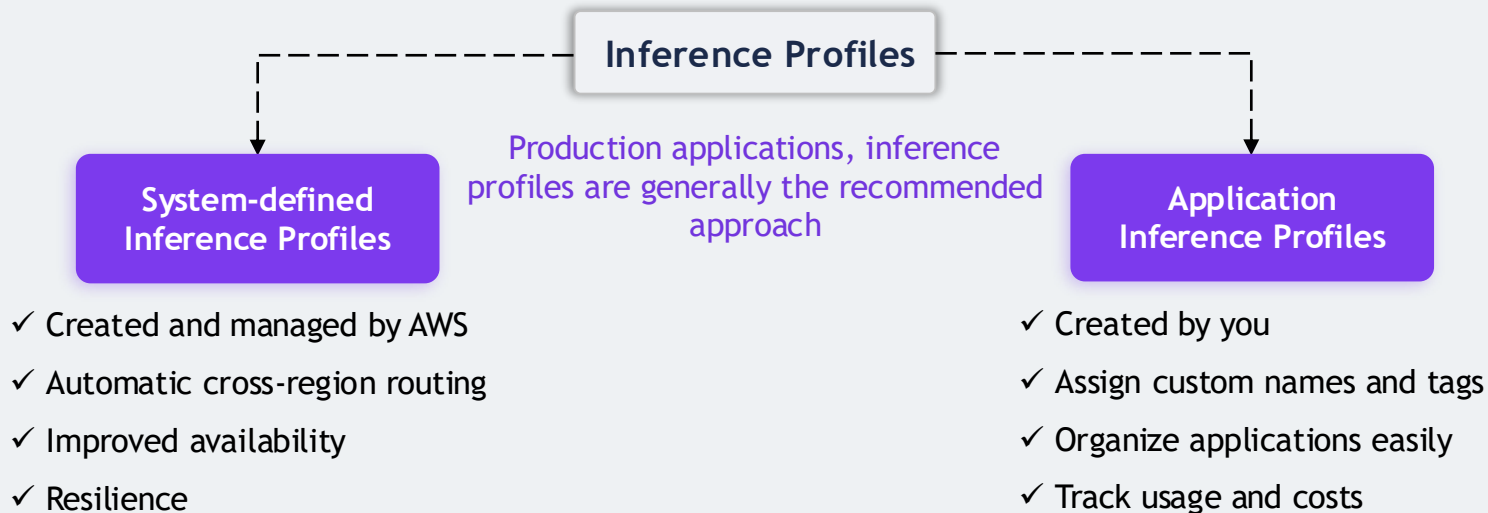
Centralizes model management

Simplifies application configuration

Organizes model usage

Tracks usage across teams

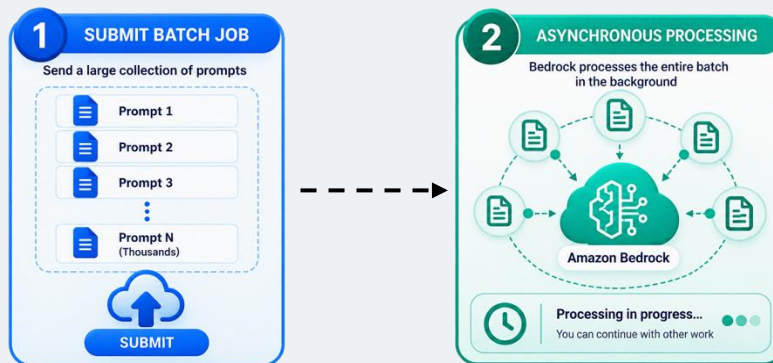
What are Inference Profiles?



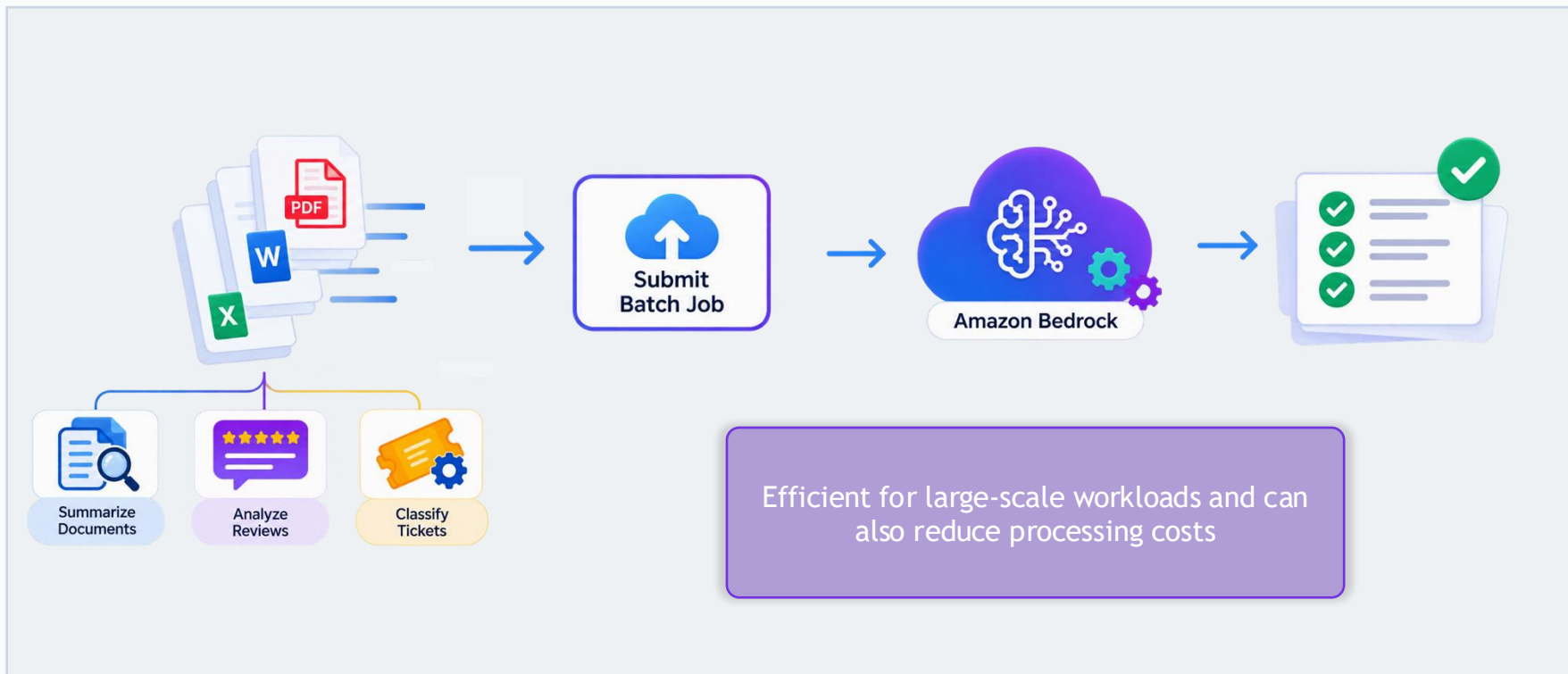
What is Batch Inference?

Batch Inference

Instead of processing one request
at a time



What is Batch Inference?



Inference Profiles vs Batch Inference

Managing production
workloads

**Inference
Profiles**

Manage how your application
accesses foundation models

Large-scale offline
processing

**Batch
Inference**

How large volumes of requests
are processed efficiently

Model Selection Guide

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What is Amazon Bedrock

GPU Infrastructure

**Machine
learning expertise**

Large AI models

What is Amazon Bedrock

GPU Infrastructure

Machine
learning expertise

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Machine
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Scaling mechanisms

Machine
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Large AI models

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APIs

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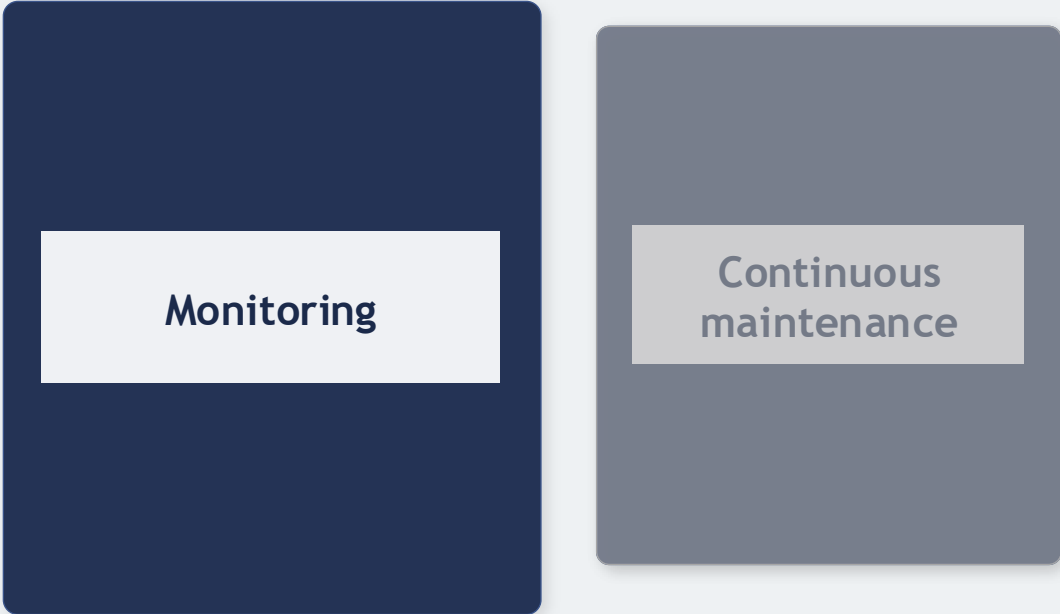
What is Amazon Bedrock

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Security

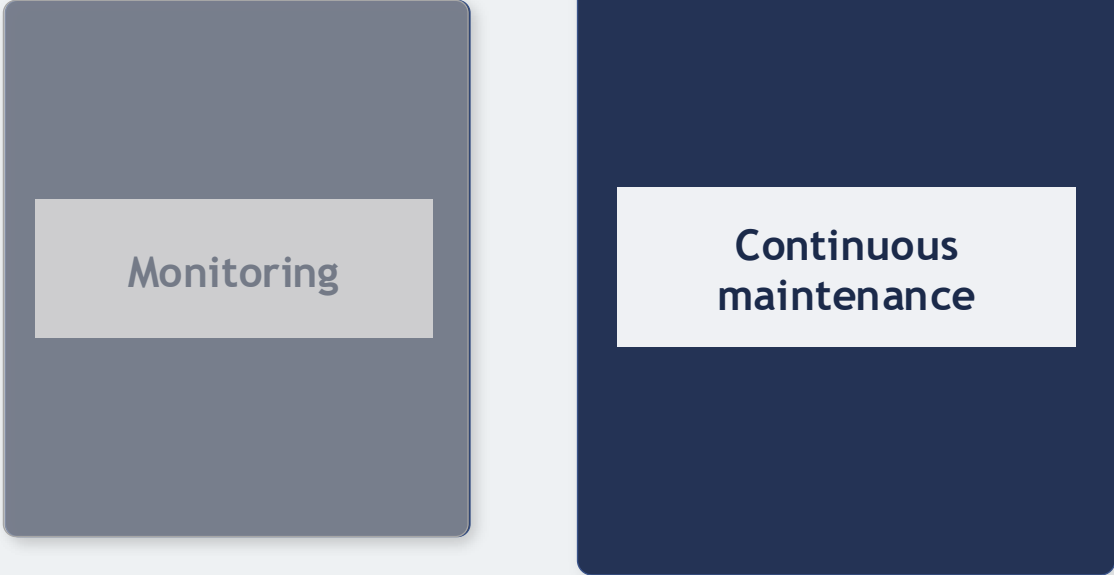
What is Amazon Bedrock



Monitoring

Continuous
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- ✓ Managing servers
- ✓ Building infrastructure



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Unified gateway to
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AWS service

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stability.ai

Stability AI



Meta



Amazon Titan
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Customer support chatbot

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GPU-based
servers



Large language
model



Configure
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Manage
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Optimize
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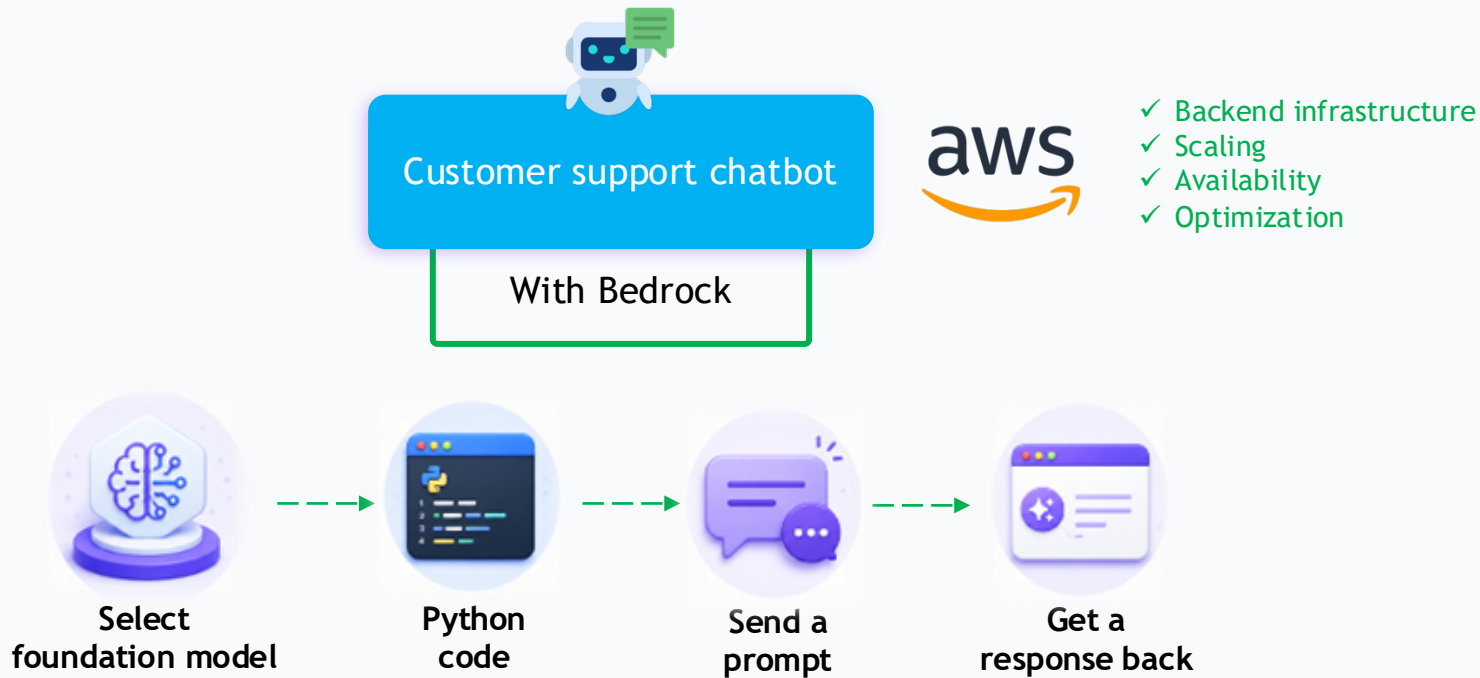


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Continuously
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What is Amazon Bedrock



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reasoning tasks



Open-source
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- ✓ Rewriting parts of your application

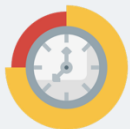
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Without Managed Platform



Buy hardware

Configure wiring

Install software

Troubleshoot issues

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With Managed Platform

Everything is already integrated
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Works the same way for
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Handles the infrastructure
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Key Features & Functionalities

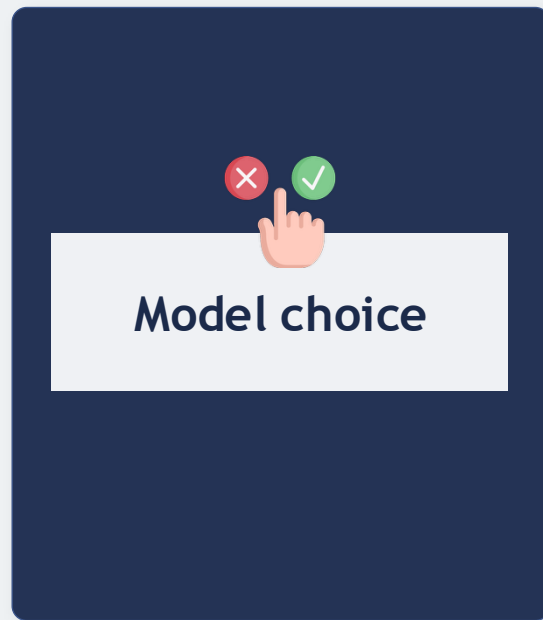
Section 1 › Introduction to Amazon Bedrock

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Key Features



Key Features



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Cost Optimization

Key Features



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Databases



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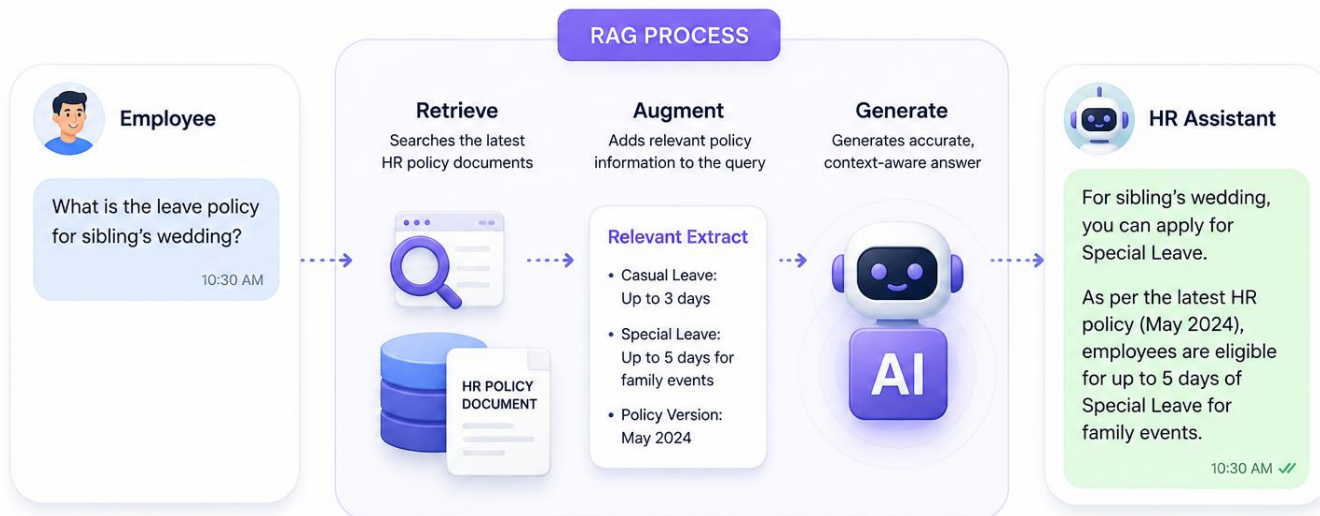
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Key Features



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Data automation

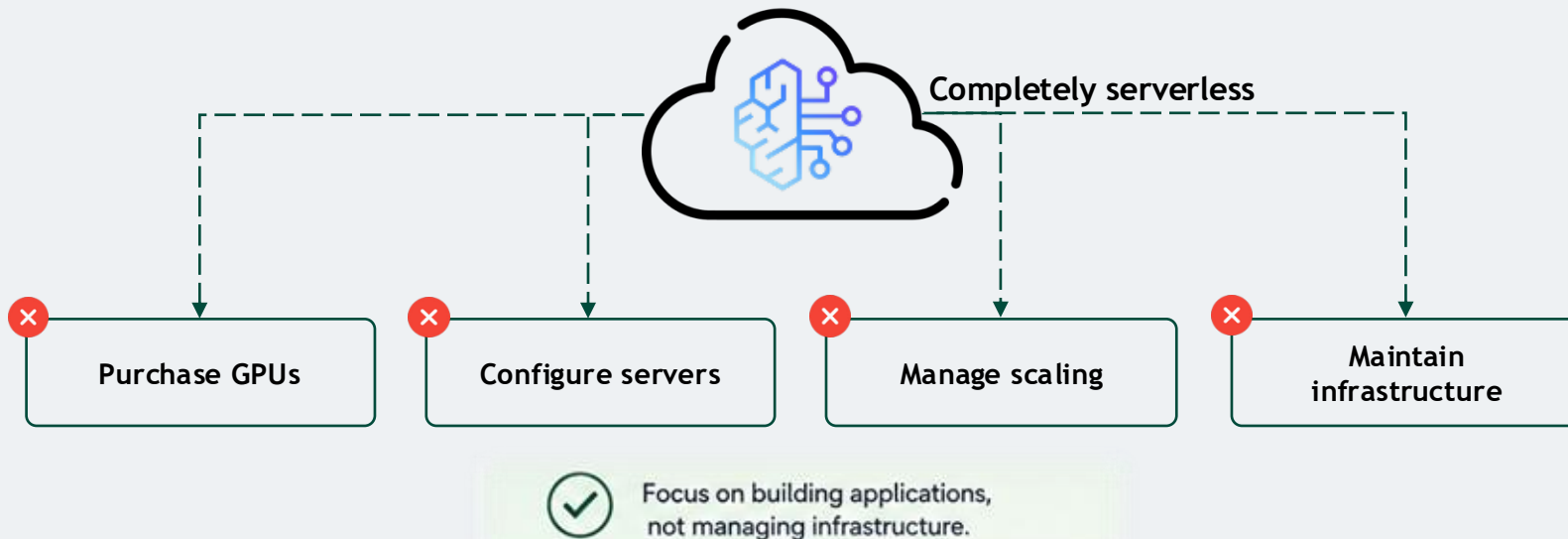
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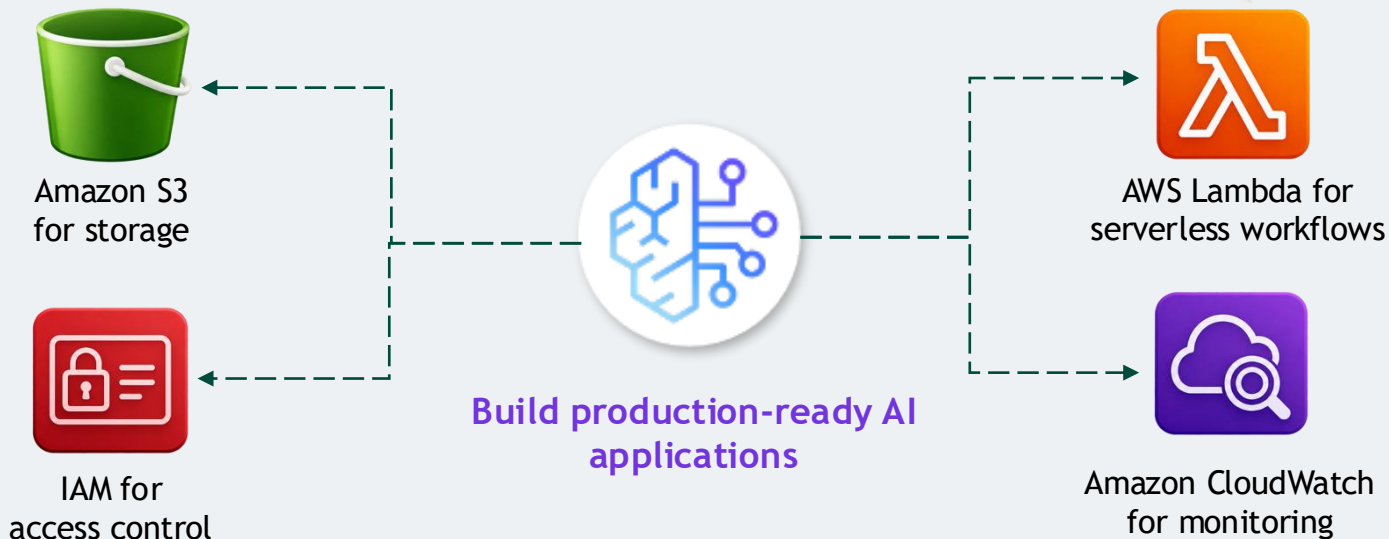


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Section 1 › Introduction to Amazon Bedrock

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Create an
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2



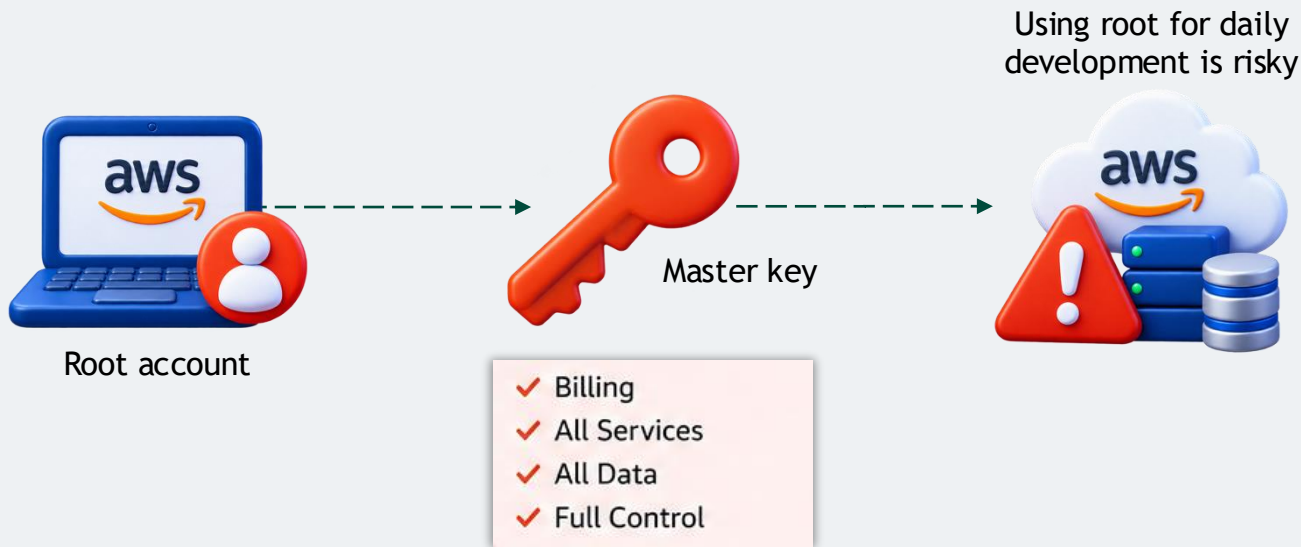
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access keys

3

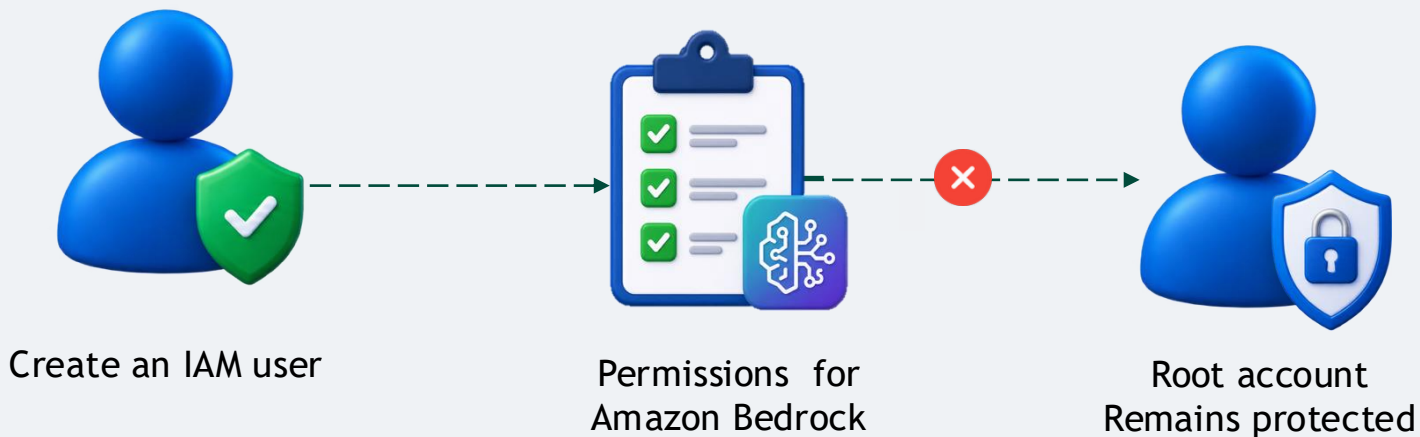


Install & configure
the AWS CLI

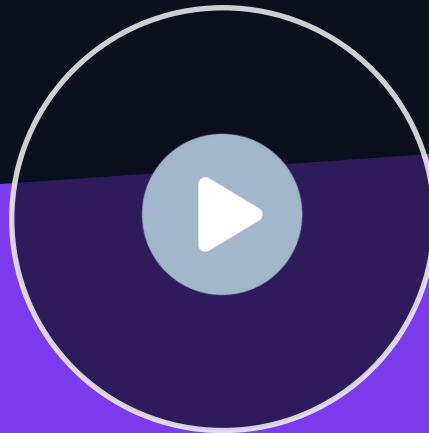
Understanding Why IAM Matters



Understanding Why IAM Matters



L I V E D E M O



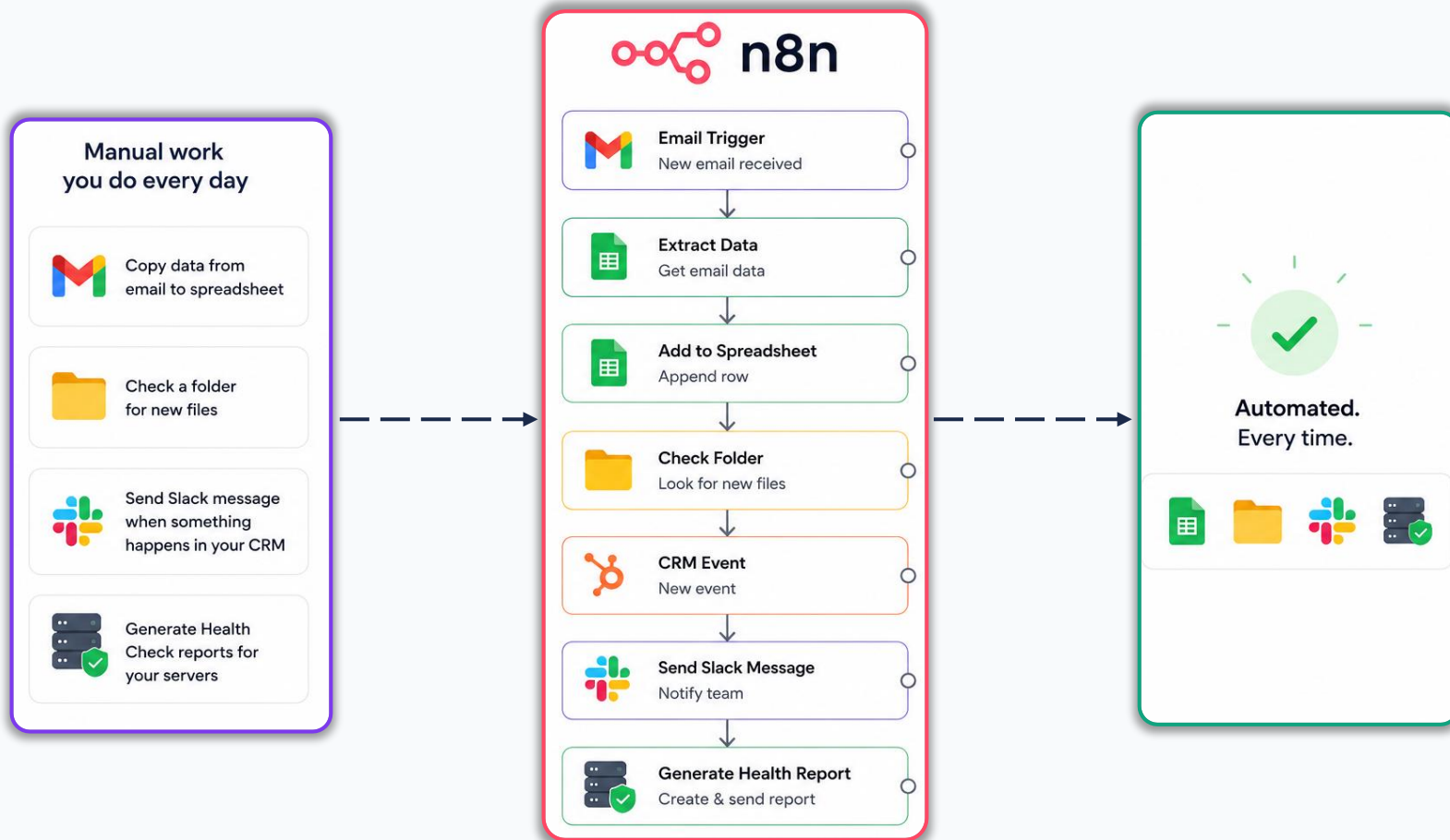
Pre-requisites Setup (AWS console & CLI)

Use cases & Industry Applications

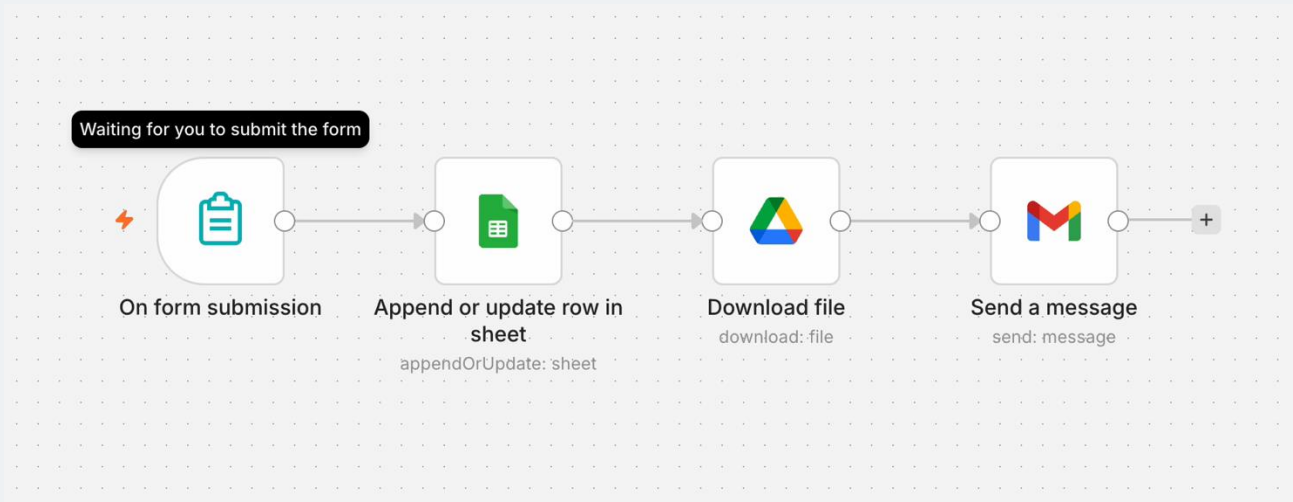


**Where can we actually use
Amazon Bedrock?**

Use Cases & Industry Applications



“Workflow Automation, the Low-Code/No-Code Way”



Drop into JavaScript or Python code when
you need more control

“Workflow Automation, the Low-Code/No-Code Way”



Open source, released
under a fair-code license



“Workflow Automation, the Low-Code/No-Code Way”



“Workflow Automation, the Low-Code/No-Code Way”



“Workflow Automation, the Low-Code/No-Code Way”



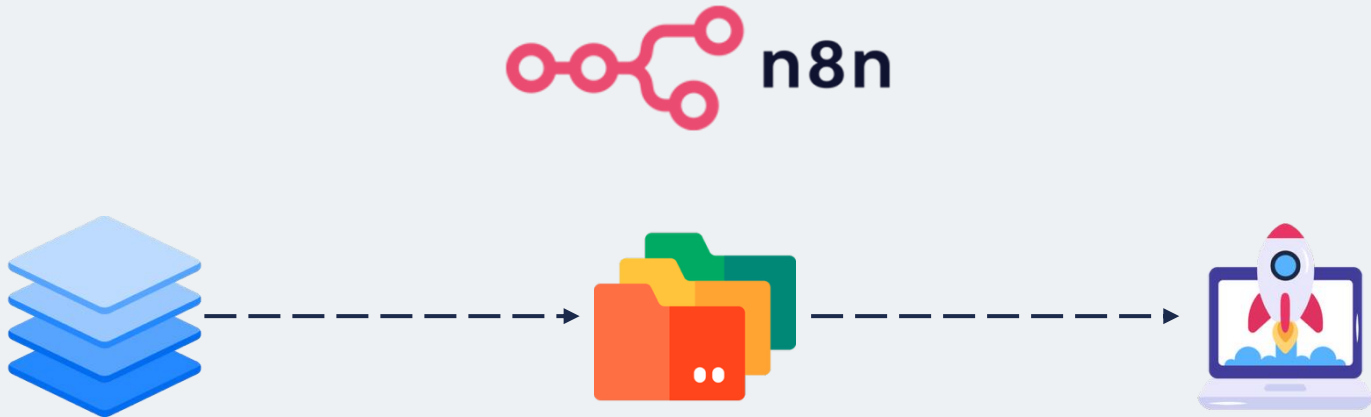
Gives you full code access when visual nodes aren't enough

Run entirely on your own infrastructure

Data privacy or cost control matters to you

Bills you per workflow execution, not per individual step

“Workflow Automation, the Low-Code/No-Code Way”



AI-powered agents, SRE automation, and end-to-end business workflows

“Workflow Automation, the Low-Code/No-Code Way”

- ✓ n8n is a visual, node-based workflow automation
- ✓ It is a Low-code/no-code tool which is Open source, fair-code licensed
- ✓ Execution-based billing, not per-step
- ✓ Increasingly the go-to tool for AI agent workflows

Available n8n Setup Offerings

Section 1 > Foundations of AI Agents



n8n Cloud



n8n On-Prem



n8n Cloud

n8n's fully managed SaaS offering Don't manage any servers — n8n hosts everything, handles updates, scaling, and uptime

STARTER

Build and automate
your workflows



PRO

Scale and automate
without limits



STARTER

Build and automate
your workflows



PRO

Scale and automate
without limits



Scale based on how many workflow executions you need per month

Team members can collaborate, and how much execution history is retained

Cloud is the right choice when you want to get moving fast, without touching infrastructure at all

On-Prem (Self-Hosted)



Own server

Own cloud VM

Own data center

Gives you full control over your data, which matters a lot for companies with compliance or data-residency requirements

Community Edition VS Licensed Editions

Community Edition

Free open-source version of n8n

Install it and run it with no license fees and no execution limits

Includes the full node library and core functionality

Things you don't get here are enterprise-grade extras

Single sign-on, advanced role-based access control, and dedicated support

Licensed Editions

With a license key on your own instance

Business

Enterprise

Community Edition VS Licensed Editions

- ✓ n8n is a visualJust exploring or learning → Community Edition, self-hosted. Zero cost, full functionality
- ✓ Want zero setup, willing to pay → n8n Cloud, Starter or Pro.
- ✓ Team needs SSO, audit trails, or dedicated support, but wants to self-host → Business or Enterprise license
- ✓ Large org with compliance and support requirements → Enterprise, self-hosted or cloud

A Quick recap:

- N8n comes with Two models: n8n Cloud (managed SaaS) and On-Prem (self-hosted)
- On-Prem splits into Community Edition (which is free and open source) and Licensed (with two offerings Business & Enterprise – via license key)
- Cloud offers Starter and Pro offerings as its managed plans
- For this course, we'll self-host our n8n using Community Edition

SECTION INTRODUCTION

Prompting Techniques and Prompt Management

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 4 — Prompting Techniques and Prompt Management

Prompting Techniques and Prompt Management

4.0

Section Introduction

4.1

Prompt Anatomy

4.2

Zero-Shot Prompting

4.3

Few-Shot Prompting

4.4

Chain of Thought (CoT) Prompting

4.5

ReAct Prompting (The Bridge to Agents)

Prompting Techniques and Prompt Management

4.0

Section Introduction

4.6

Role-Based Prompting

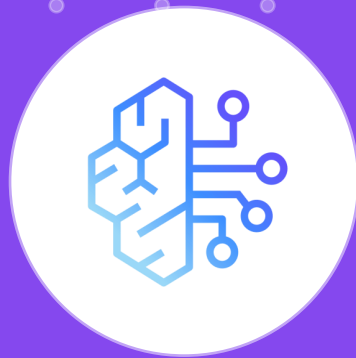
4.7

Structured Output Prompting

SECTION INTRODUCTION

Build with Amazon Bedrock

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 5 — Build with Amazon Bedrock.

Build with Amazon Bedrock.

5.0

Section Introduction

5.1

Token & Tokenization

5.2

Tokenizers

5.3

Prompt Testing & Optimization

5.4

Understanding Embeddings and Vector Databases

5.5

Document Chunking, Embeddings, and Semantic Search

Build with Amazon Bedrock.

5.0

Section Introduction

5.6

Knowledge Base and RAG Overview

5.7

Demonstration - Building Your Own Knowledge Base

Knowledge Base & RAG Overview



Section 5 › Build with Amazon Bedrock

Knowledge Base & RAG Overview

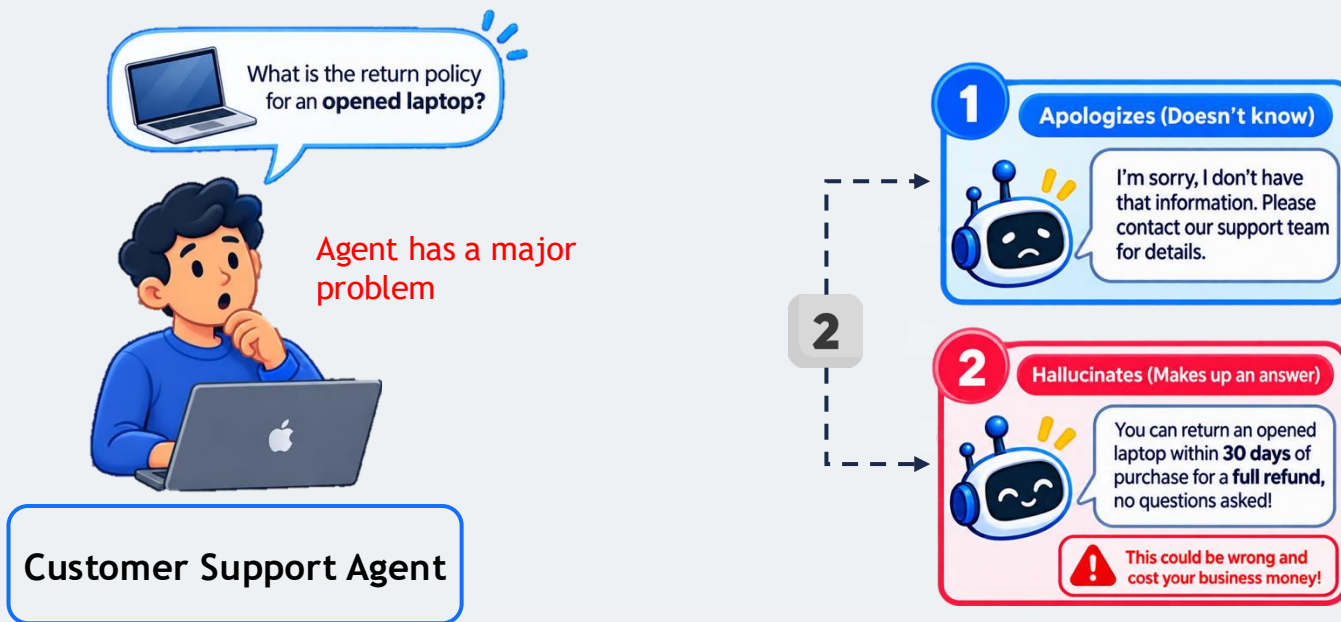


Prompt Management in the Bedrock console



Learning how to properly instruct,
save, & manage our model interactions

Knowledge Base & RAG Overview



Knowledge Base & RAG Overview

Foundation
Models

- ✓ Trained on massive, general datasets
- ✓ Up to a specific point in time

Proprietary company data

Knowledge Base & RAG Overview



RAG

Retrieval Augmented Generation

Knowledge Base & RAG Overview

RAG

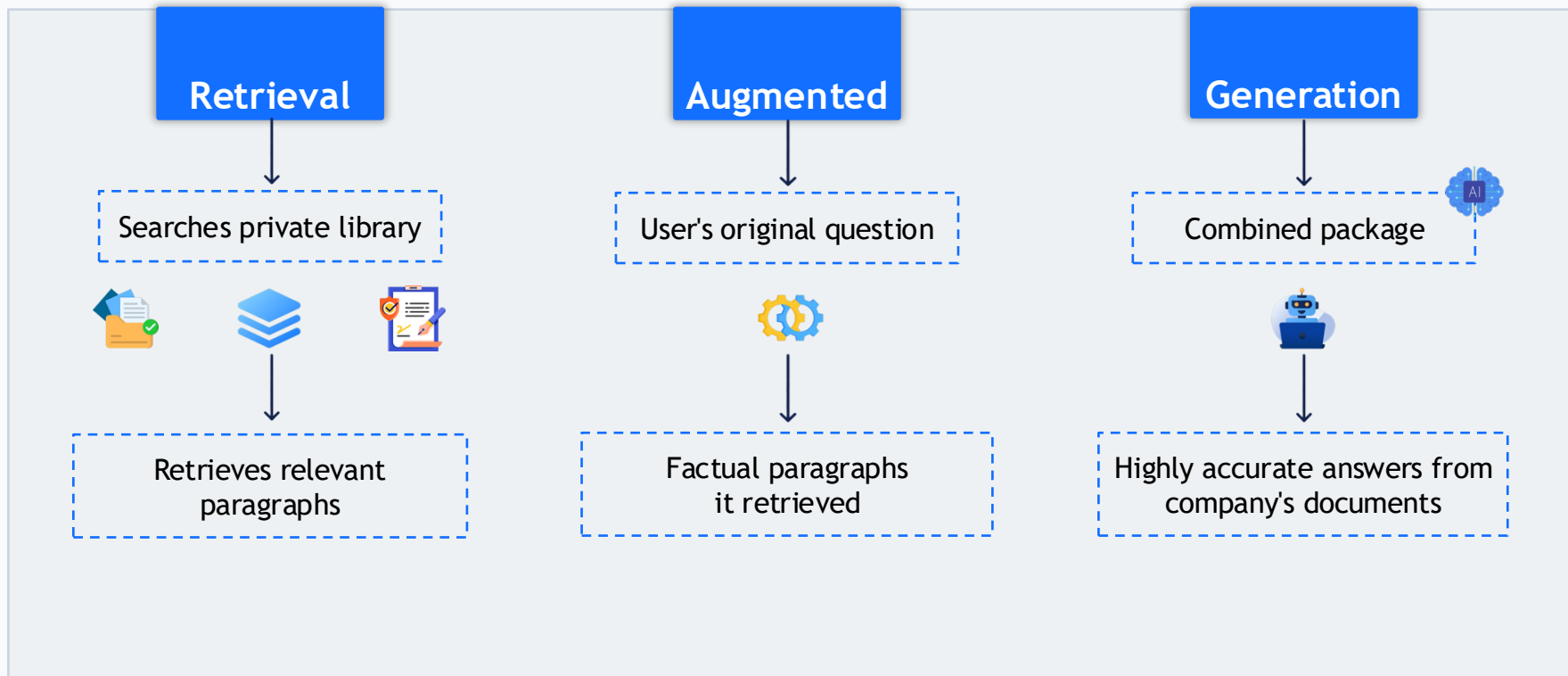
Acts like an open book



Highly specific question

Open Book test

Knowledge Base & RAG Overview



Knowledge Base & RAG Overview



Handles the end-to-end orchestration for you

Knowledge Bases

Completely automates the heavy lifting of RAG

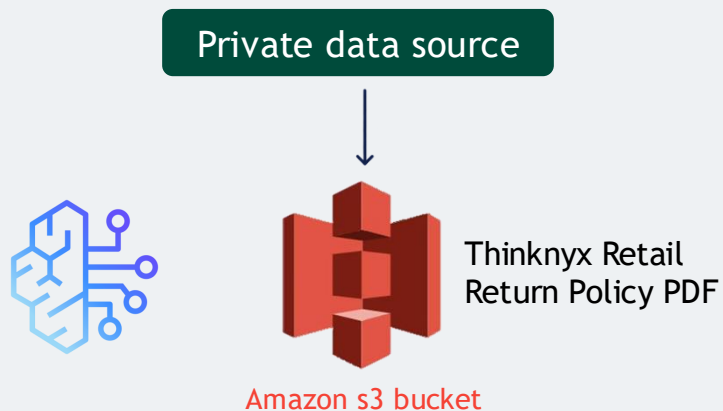


Custom code

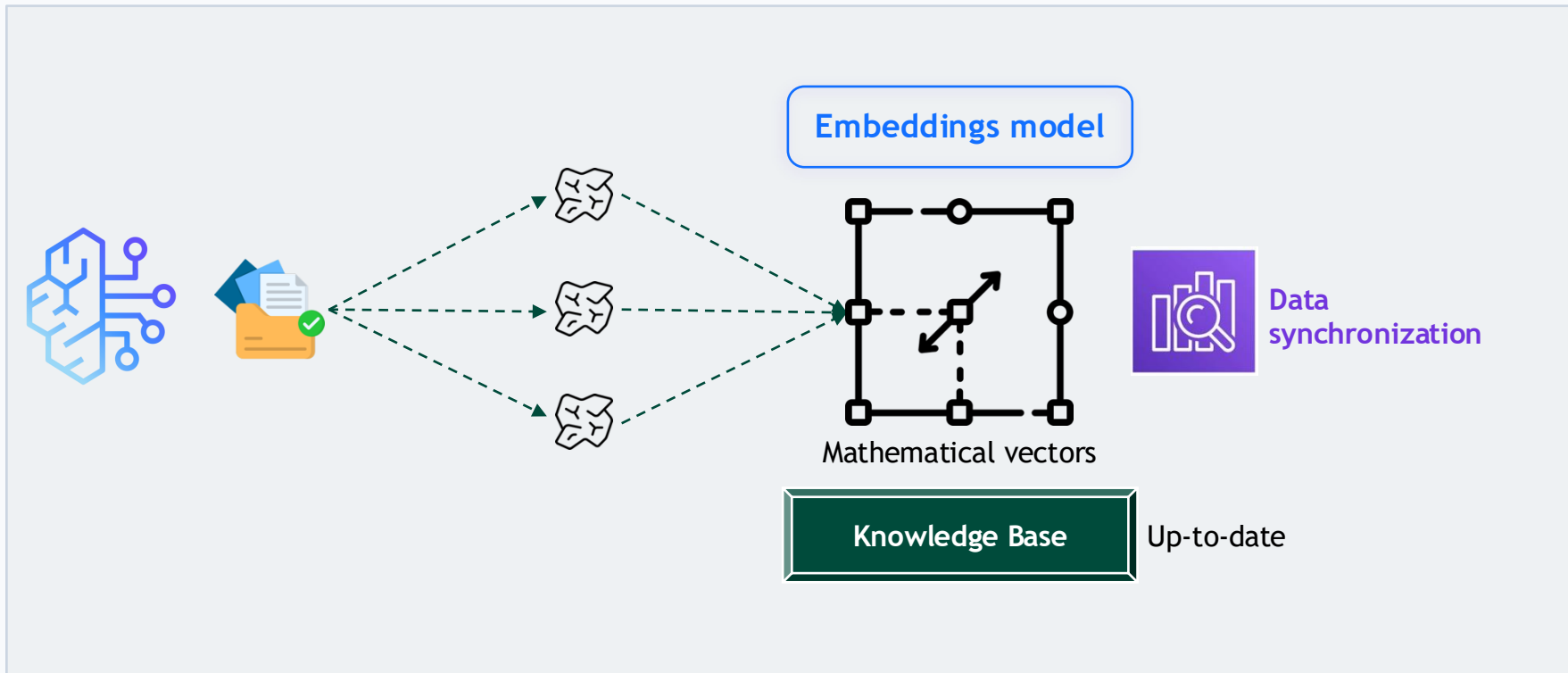
Building complex ETL pipelines

Manually setting up databases

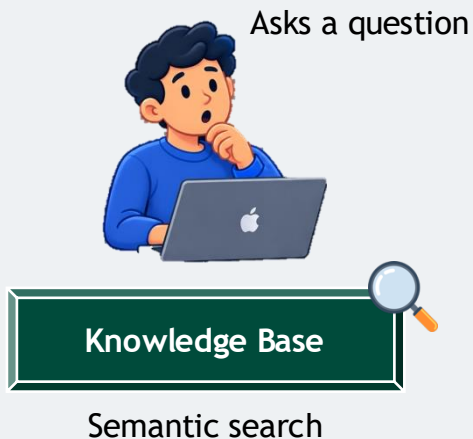
Knowledge Base & RAG Overview



Knowledge Base & RAG Overview

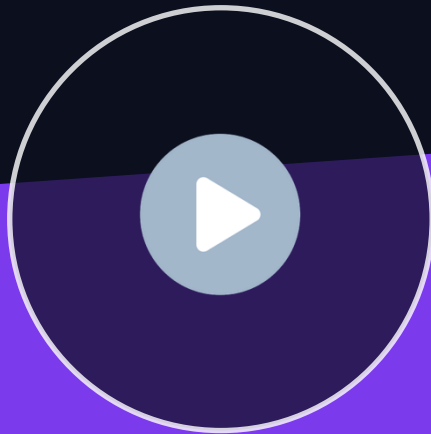


Knowledge Base & RAG Overview



- ✓ Instantly retrieve the most relevant information
- ✓ Feeds it as context to the Foundation Model
- ✓ Delivers a precise, factual answer

L I V E D E M O

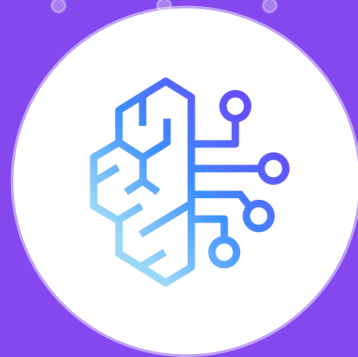


Explore Flow Builder

SECTION INTRODUCTION

Build with Flows

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 6 – Build with Flows

Build with Flows

6.0

Section Introduction

6.1

Overview of No-Code & Low-Code

6.2

Explore Flow Builder

6.3

Flows with Prompt & Knowledge Base Nodes

6.4

Implementing flows with Condition Node

6.5

Implementing flows with Do While Loop

Build with Flows

6.0

Section Introduction

6.6

Implementing flows with Iterator node

6.7

Implementing flows with Inline Code

6.8

Implementing flows with Lambda Function

Overview of No-Code & Low-Code



Section 6 › Build with Flows

What are No-Code and Low-Code?



Building an application or workflow using visual components instead of writing programming code

What are No-Code and Low-Code?



- ✓ Collect customer information
- ✓ Validate it
- ✓ Send an email



Low-code

What are No-Code and Low-Code?

No-code

Build primarily through
visual components

Low-code

Build visually, with the option to
add code when required



Build and modify applications faster with less manual coding

Example of No-Code and Low-Code

Workflow

Traditional Development

Write code for each step and handle integrations



Amazon Bedrock Flows

Drag, connect, and create visually

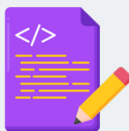


What are Amazon Bedrock Flows?



**Amazon
Bedrock Flows**

Visual environment for building AI workflows by connecting different components on a canvas



- ✓ Connect a prompt
- ✓ Knowledge Base
- ✓ Lambda function
- ✓ Other components

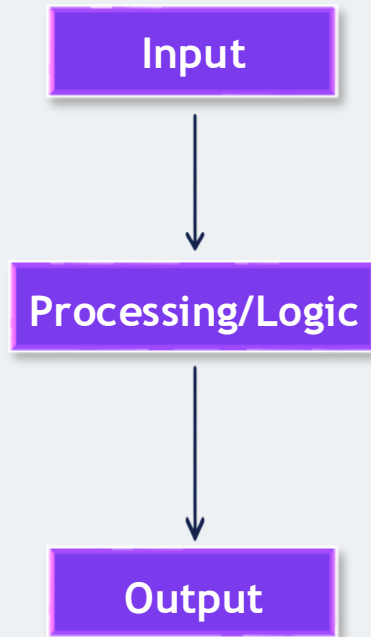


Connect them
using visual links

Bedrock Flows Architecture

- **Input components :-** Bring information into the workflow
- **Processing components :-** Work on that information
 - Prompts
 - Knowledge Bases
 - Lambda functions
- **Logic components :-** Control how the workflow behaves, such as
 - conditions
 - Loops
 - Iterators
- **Output components :-** Returns the result

Bedrock Flows Architecture



Can see this entire workflow visually on a single canvas

Key Advantages of Bedrock Flows

Speed

Collaboration

Easy Iteration

Key Advantages of Bedrock Flows

Speed

Collaboration

Easy Iteration

Key Advantages of Bedrock Flows

Speed

Collaboration

Easy Iteration

Key Advantages of Bedrock Flows

Built-in Testing

Serverless

Key Advantages of Bedrock Flows

Built-in Testing

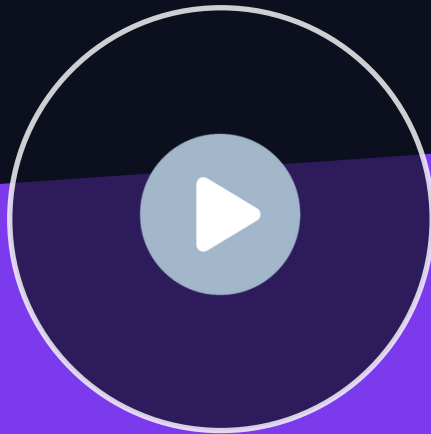
Serverless

Real-World Example



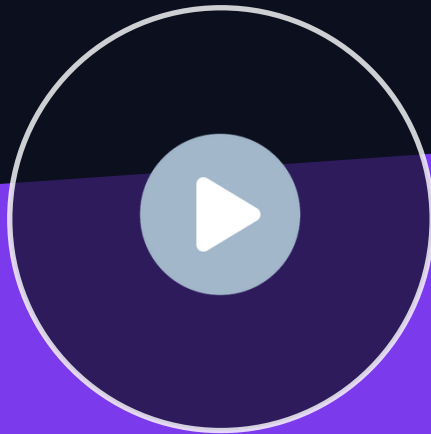
Instead of implementing every connection through Python code, we can visually build this workflow using Bedrock Flows

L I V E D E M O



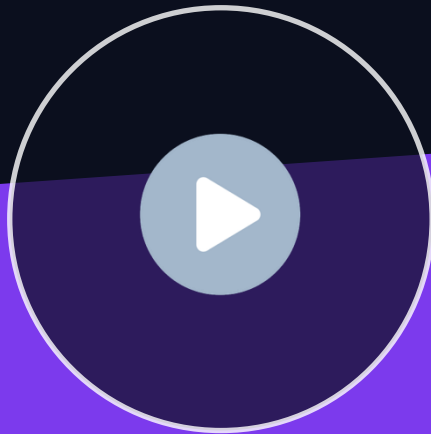
Explore Flow Builder

L I V E D E M O



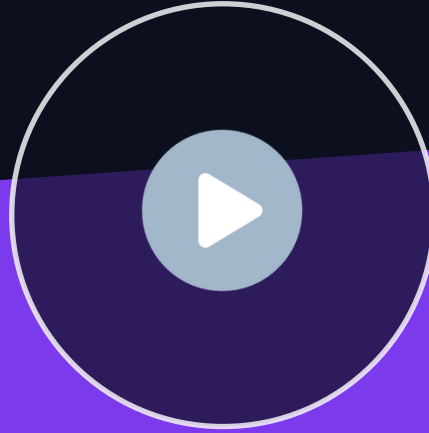
Flows with Prompt

LIVE DEMO



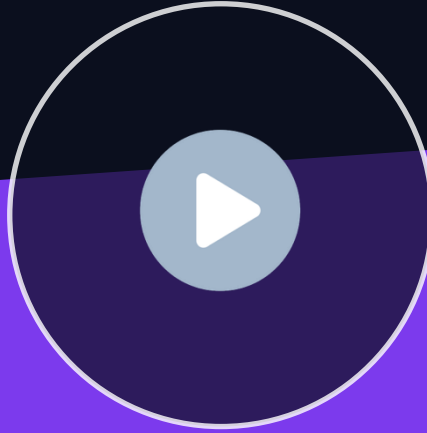
Flows with Knowledge Base

L I V E D E M O



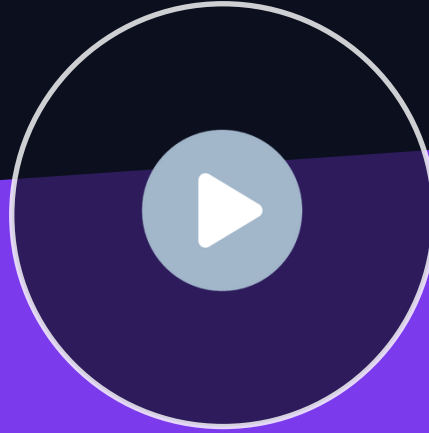
Implementing flows with Condition Node

L I V E D E M O



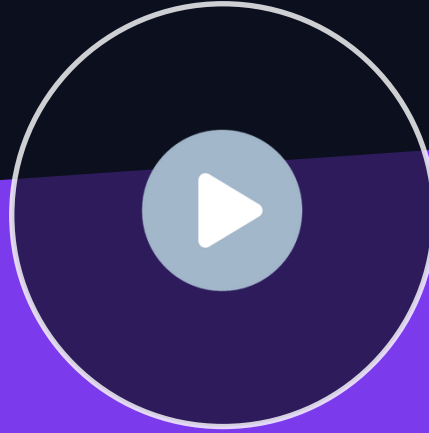
Implementing flows with Do While Loop

L I V E D E M O



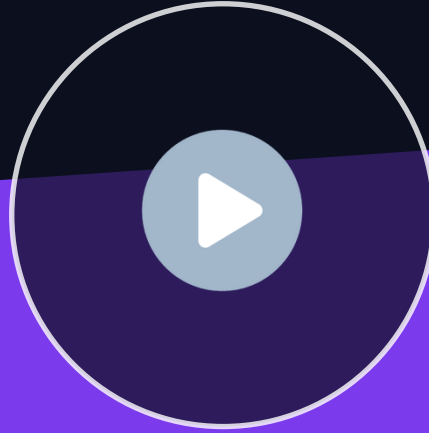
Implementing flows with Iterator node

L I V E D E M O



Implementing flows with Inline Code

LIVE DEMO

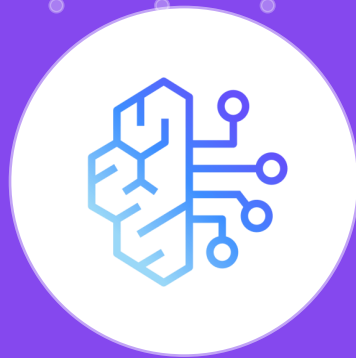


Implementing flows with Lambda Function

SECTION INTRODUCTION

Advanced Bedrock Functionalities

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 7 — Advanced Bedrock Functionalities

Advanced Bedrock Functionalities

7.0

Section Introduction

7.1

Introduction to AgentCore

7.2

Building the AgentCore Harness & Knowledge Gateway

7.3

Integrating AgentCore into an Interactive CLI

7.4

Bedrock's Security & Guardrails

7.5

Creating a Guardrail & Applying in Flows

Advanced Bedrock Functionalities

7.0

Section Introduction

7.6

Fine-Tuning & Model Customization

7.7

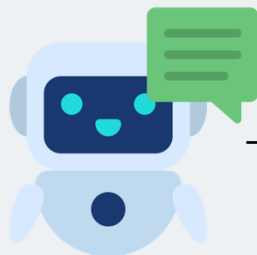
Cost, Latency, & Optimization

Introduction to AgentCore

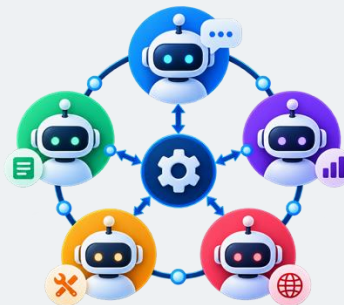
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Section 7 › Advanced Bedrock Functionalities

Introduction to AgentCore



**Simple
chatbot**



**Complex
multi-agent systems**

Introduction to AgentCore



**Amazon
Bedrock AgentCore**

Introduction to AgentCore

Organizations prefer to build their agents
using open-source frameworks



LangGraph



LlamaIndex

Handling identity

Security

Hosting

Browser integrations

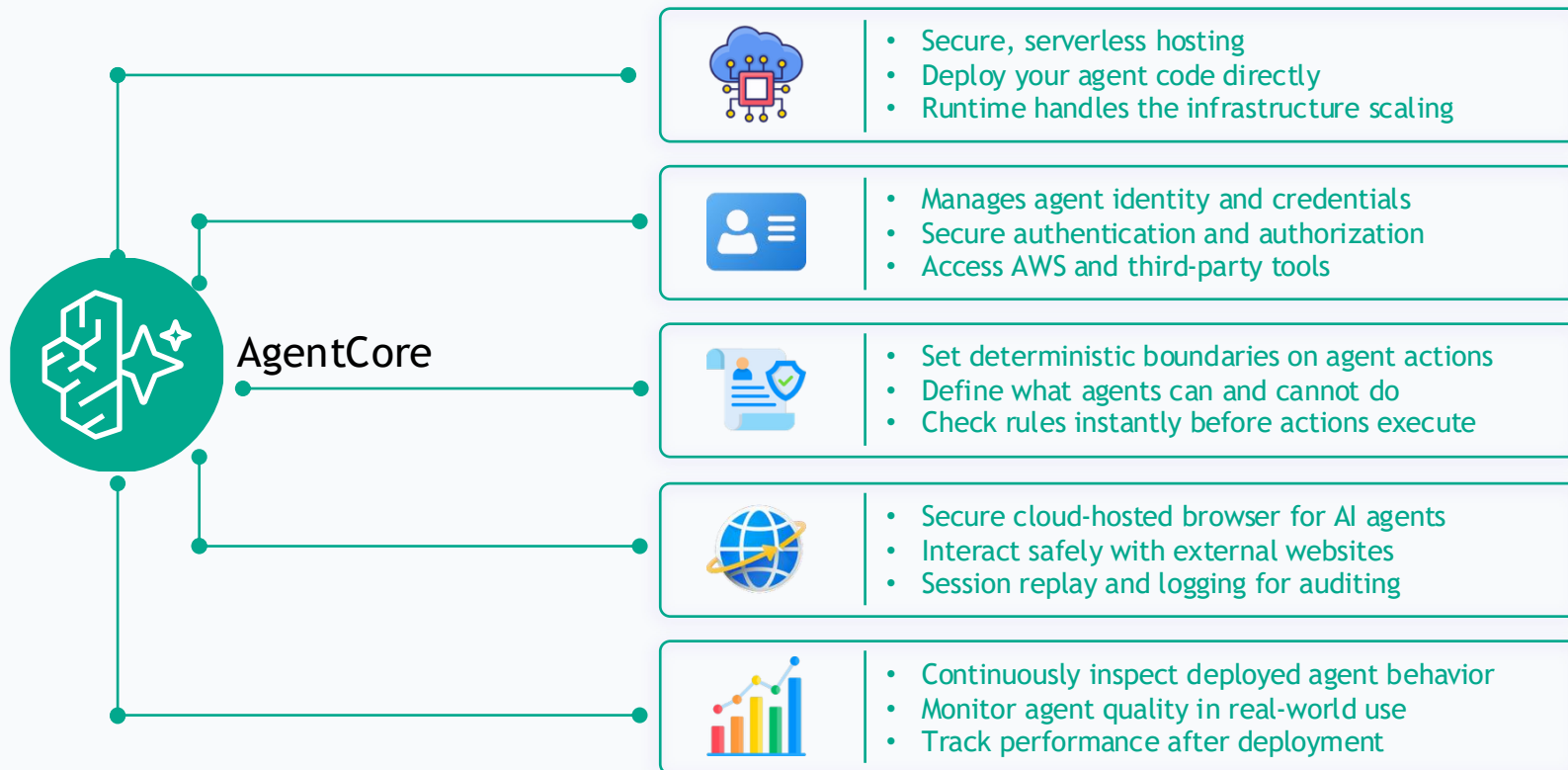
Introduction to AgentCore



AgentCore

- ✓ Providing purpose-built infrastructure design
- ✓ Specifically for dynamic AI agents
- ✓ Regardless of the framework or model you choose to use

Introduction to AgentCore

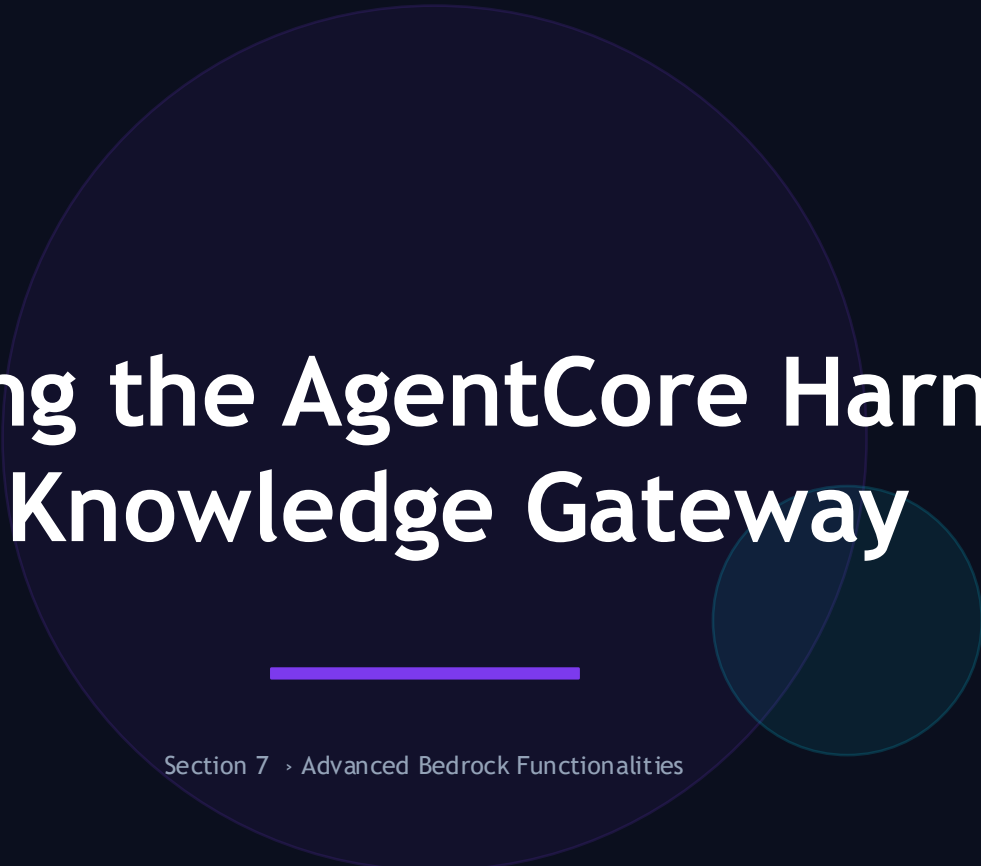


Introduction to AgentCore



Enterprise operating system for your AI agents that allows developers to use their preferred open-source tools while relying on AWS for security, scale, and governance

Building the AgentCore Harness & Knowledge Gateway

The background features a large, dark purple circle on the left and a smaller, teal circle on the right, both with thin outlines. A solid purple horizontal line is positioned below the main title.

Section 7 › Advanced Bedrock Functionalities

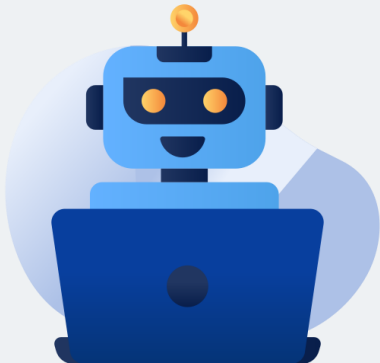


What is AgentCore?

What exactly is an Agent Harness?

Understanding AgentCore & Agent Harness

Agent Harness



Orchestration Layer → Connects the agent with the foundation model

Tool Decisions → Determines which tools to use

Memory → Remembers previous messages and context

Error Handling → Manages failures during execution

Local VS. Production → Easy to run locally, but complex at scale

Production Needs → Concurrent users, secure sandboxes, memory, identity, and scaling

Understanding AgentCore & Agent Harness

Agent Harness

Engine and environment that actually
lets your agent run safely

Understanding AgentCore & Agent Harness

Without AgentCore

Build everything from scratch



Understanding AgentCore & Agent Harness

Without AgentCore

Build everything from scratch

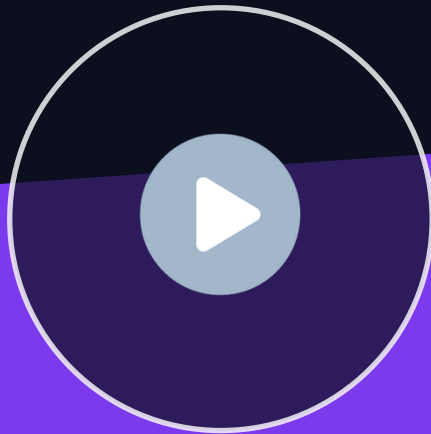


With Amazon Bedrock AgentCore

Production-ready harness, out of the box

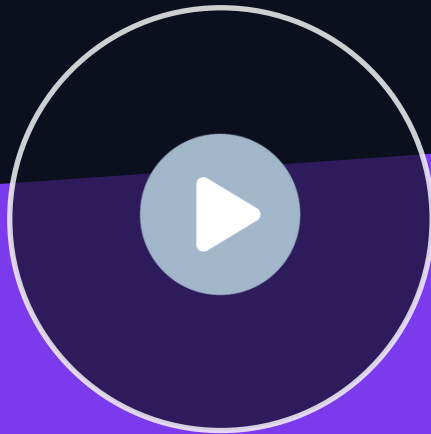


L I V E D E M O



Building the AgentCore Harness & Knowledge Gateway

L I V E D E M O



Integrating the AgentCore Gateway into an Interactive Python CLI

Bedrock's Security and Guardrails



Section 7 › Advanced Bedrock Functionalities

Bedrock's Security & Guardrails



Designed to be highly creative and helpful

Sometimes be manipulated

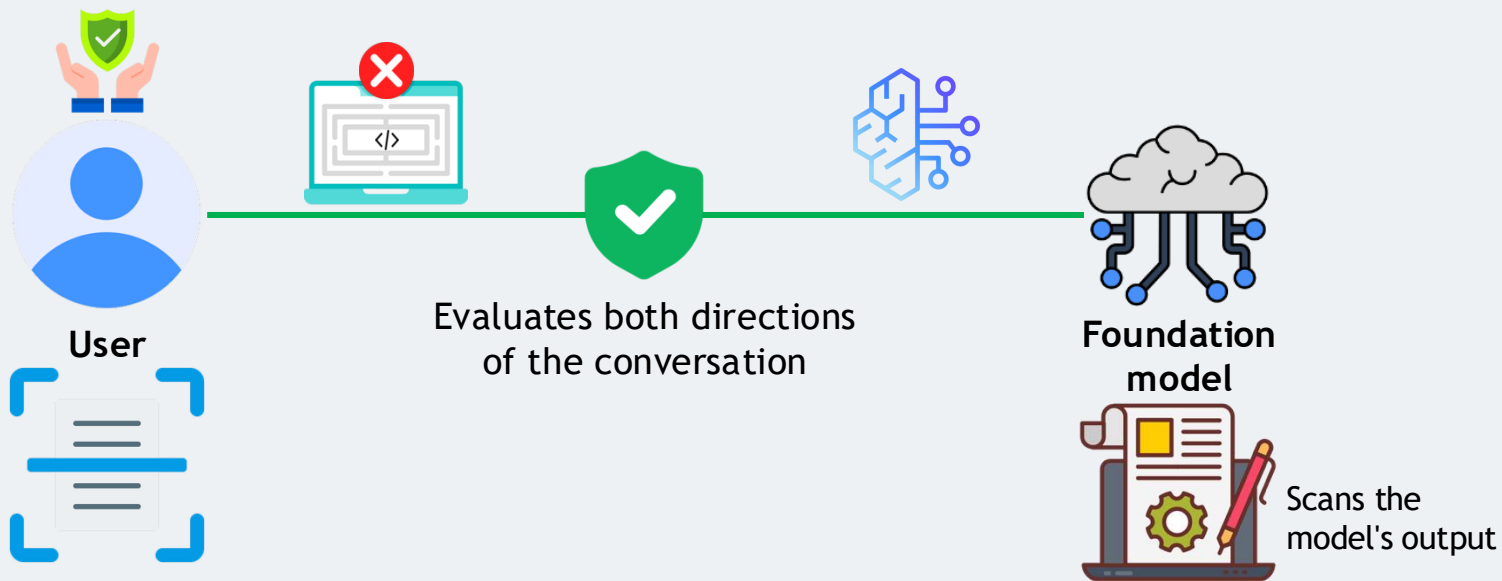
Generate inappropriate content

Leak sensitive data



Guardrails

What is an Amazon Bedrock Guardrail?



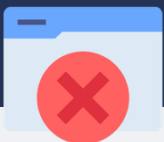
Four Protections Guardrails Provide



Content Filters

- Built-in Categories for Content Detection
- Automatically Detects Toxic Content
- Blocks Hate, Insults, Violence & Sexual Content
- Adjust Filter Strictness for Your Use Case

Four Protections Guardrails Provide



Denied Topics

- Restrict the AI to a Specific Domain
- Building a Thinknyx Retail Support agent
- Talk About Returns and Shipping
- Do Not Give Financial, Medical, or Political Advice
- Define Custom Boundaries Using Natural Language

Four Protections Guardrails Provide



Prompt Attacks

('Jailbreaking' or 'Prompt Injection')

- AI Models Are Controlled by Language
- Malicious Users Try to Trick the AI
- Examples of Prompt Injection Attacks
- Bedrock Guardrails Detect These Attacks
- Instantly Block Them Before Reaching the Foundation Model

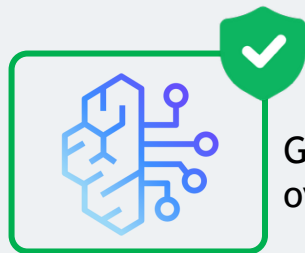
Four Protections Guardrails Provide



Sensitive Information Filters

- Detect Personally Identifiable Information (PII)
- Identify Emails, Social Security Numbers & Credit Cards
- Choose to Block the Conversation
- Or Choose to Mask the Sensitive Data
- Replace Sensitive Data with Asterisks Before the Foundation Model Sees It

To Summarize



Give us centralized control
over AI safety

Amazon Bedrock Guardrails

Filter toxic
content

Restrict
topics

Block malicious
prompt injections

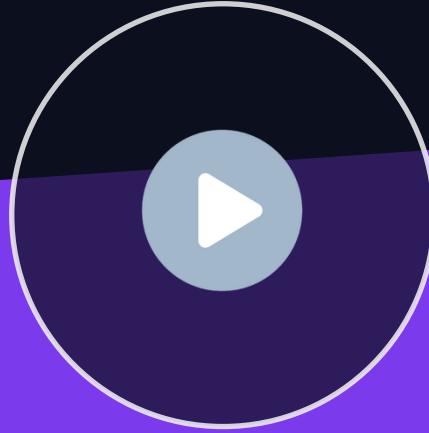
Protect user
privacy through
PII masking

To Summarize

Contextual Grounding

Helps prevent hallucinations by forcing the AI to stick strictly to your company documents

L I V E D E M O



Creating a Guardrail & Applying in Flows

Fine Tuning / Model Customization



Section 7 › First Amazon Bedrock API call

Fine Tuning / Model Customization



RAG

Inject private, factual
knowledge into foundation
models

Retrieval Augmented Generation

Fine Tuning / Model Customization

RAG

Teaches a model what to say

Fine Tuning

Modifies the model's internal weights to teach it how to say it

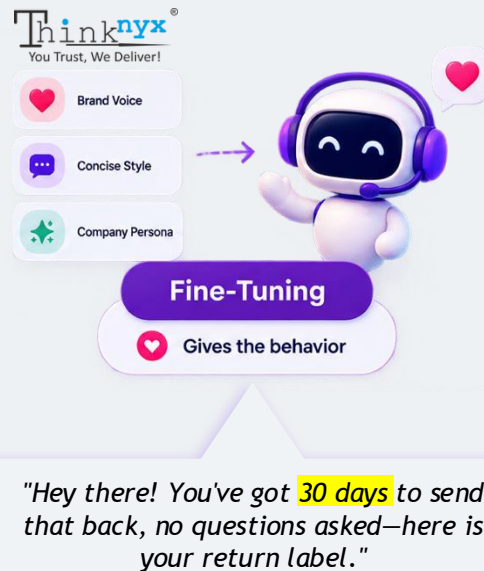
Tone

Style

Output
formatting

Domain-specific
language

Fine Tuning / Model Customization



Fine Tuning / Model Customization

Facts

RAG

Behavior

Fine Tuning

Three Primary Techniques

**Supervised
Fine-Tuning**

**Reinforcement
Fine-Tuning (RFT)**

Distillation

Three Primary Techniques

Supervised
Fine-Tuning

Reinforcement
Fine-Tuning (RFT)

Distillation

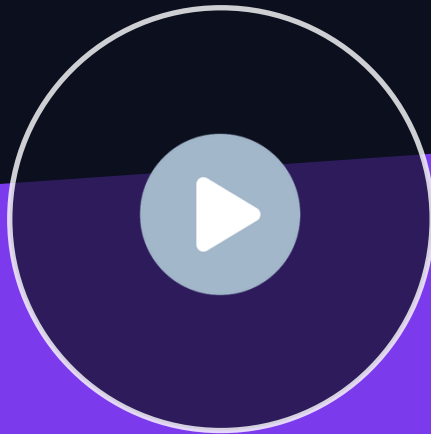
Three Primary Techniques

Supervised
Fine-Tuning

Reinforcement
Fine-Tuning (RFT)

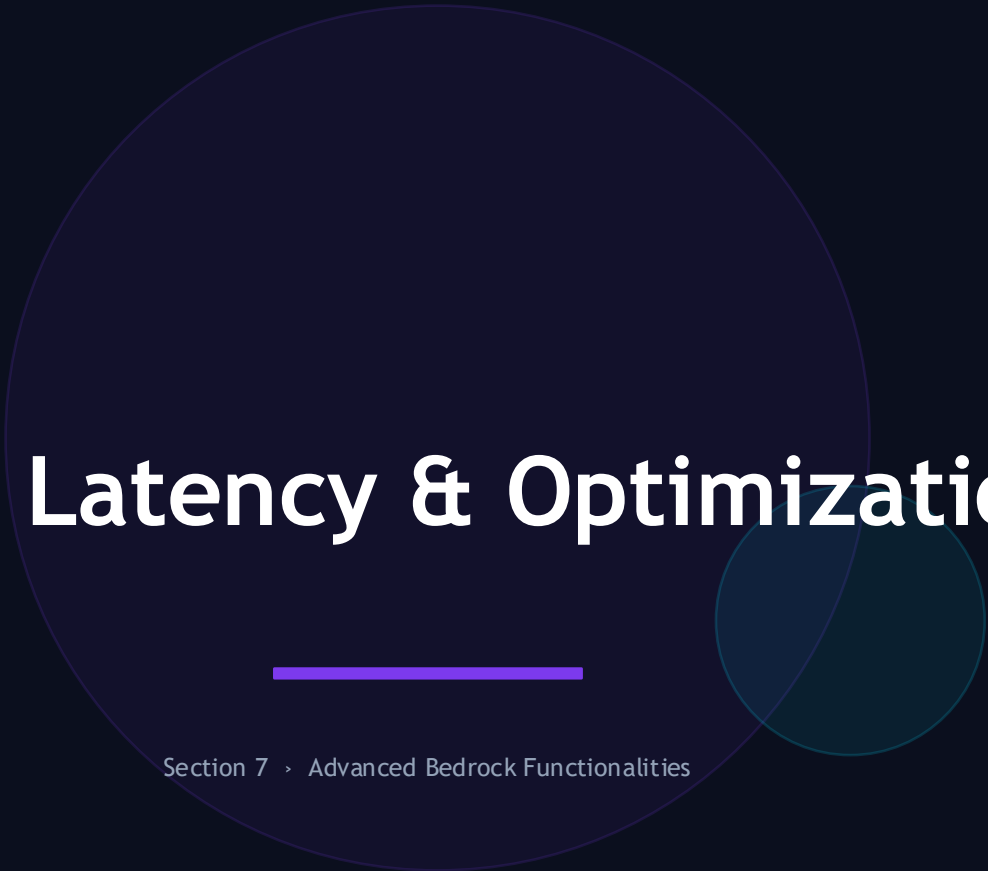
Distillation

L I V E D E M O



Fine Tuning / Model Customization

Cost, Latency & Optimization



Section 7 › Advanced Bedrock Functionalities

Cost latency & Optimization



Selecting the most powerful model for every single user request is a common mistake that will instantly inflate your budget

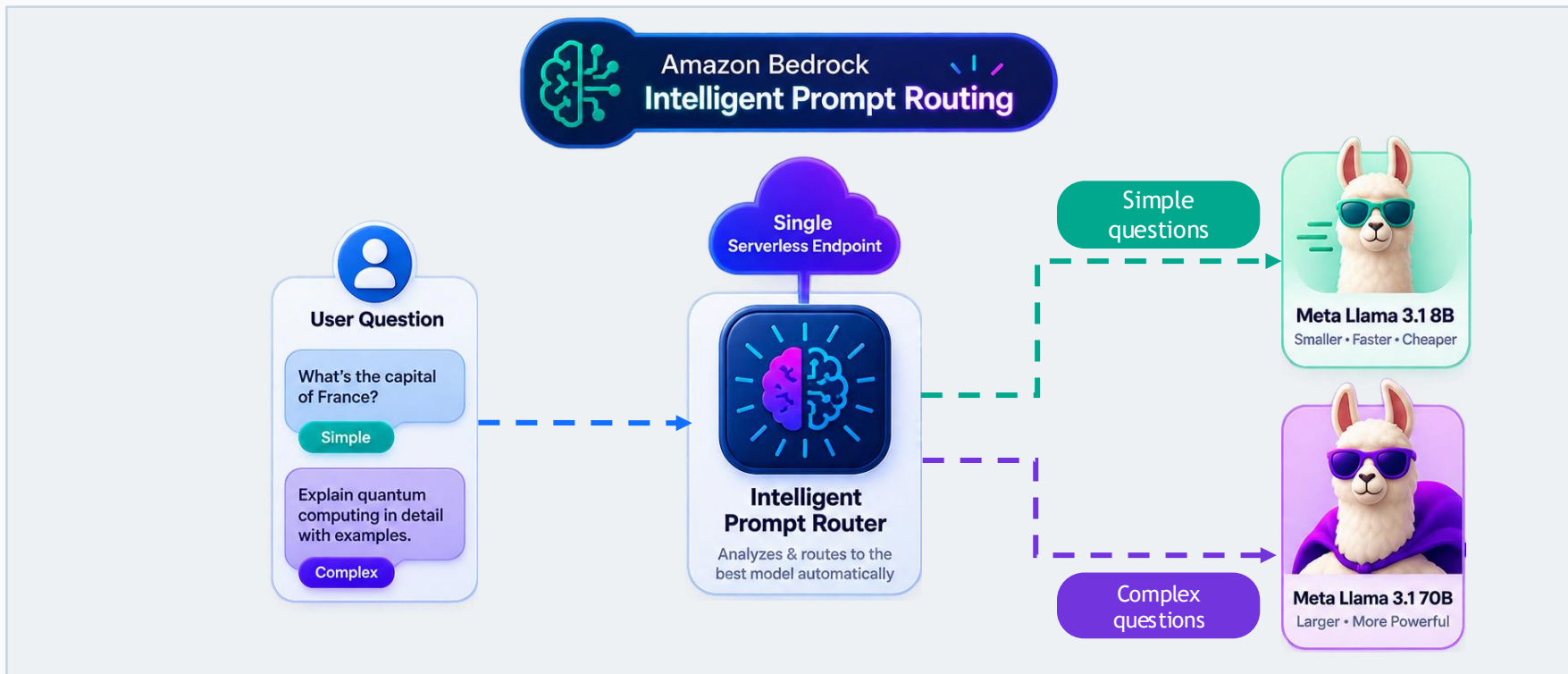
Intelligent Prompt Routing



Instead of hardcoding your app to always use a heavy, expensive model



Intelligent Prompt Routing



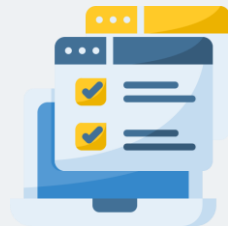
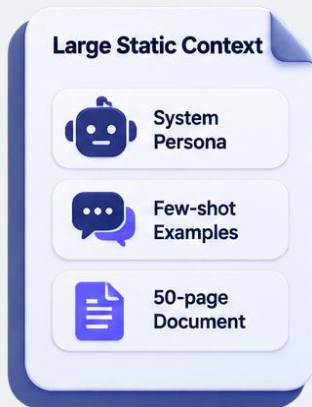
Intelligent Prompt Routing



Ensures you only pay for heavy
compute when you actually
need it

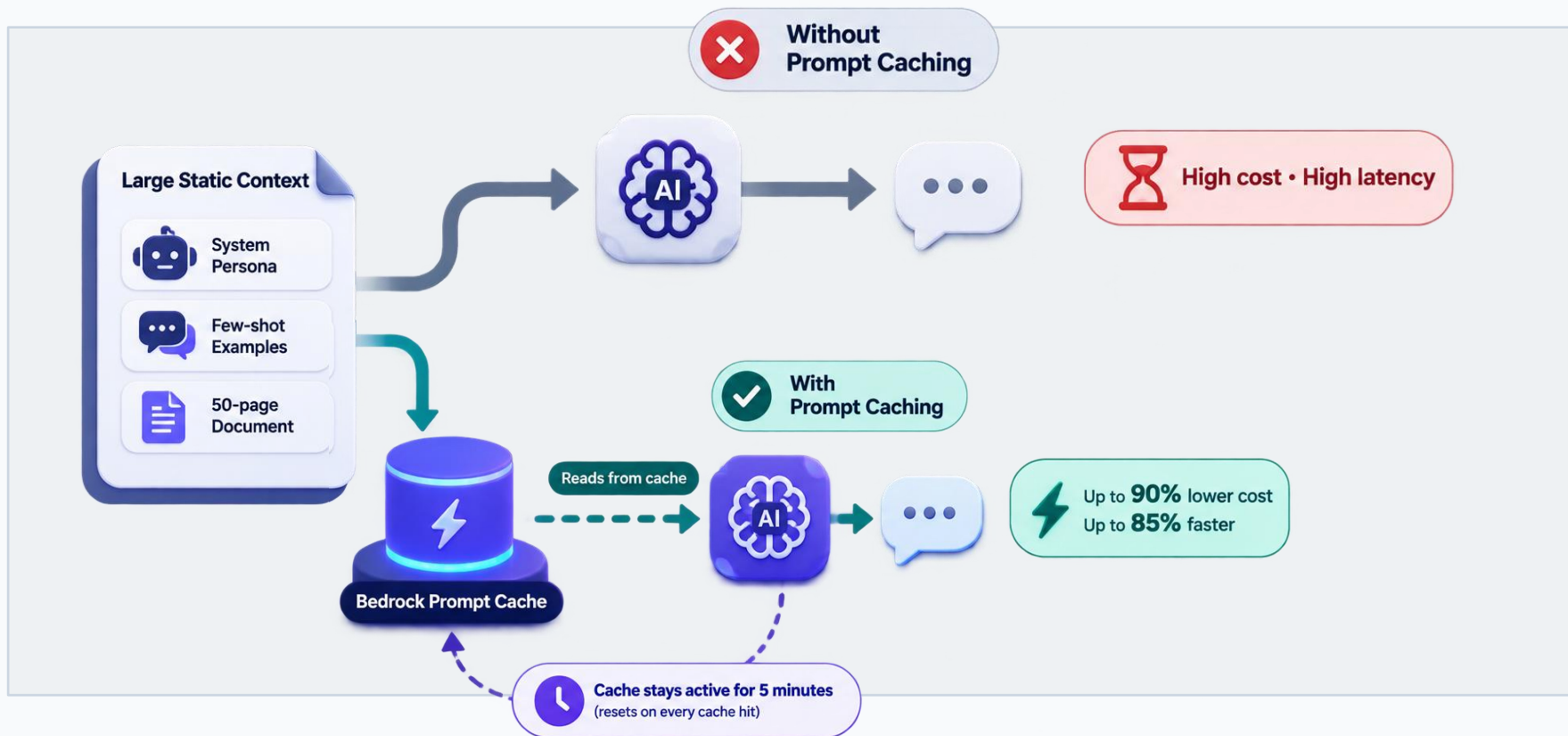
Cutting your costs
by up to 30%

Prompt Caching



Repeatedly uses a large block of static text

Prompt Caching



Execution Modes: Batch vs. Provisioned Throughput

Batch Inference

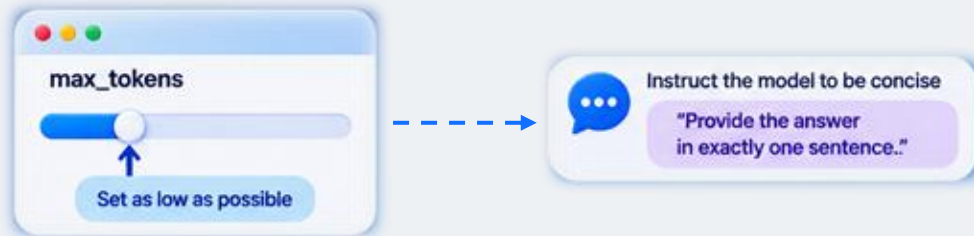
- No Real-Time Responses
- Nightly Document Summaries & Bulk Data Classification
- Submit Large Sets of Prompts to Amazon S3
- Bedrock Processes Prompts Asynchronously
- 50% Discount vs. Standard On-Demand Rates

Provisioned Throughput

- Massive, Steady Stream of Live Traffic
- On-Demand: Per-Token Costs Get Expensive
- Purchase Dedicated Model Capacity
- Fixed Hourly Rate with Provisioned Throughput
- Custom Fine-Tuned Models Require Provisioned Throughput

Output Optimization & Hidden Infrastructure

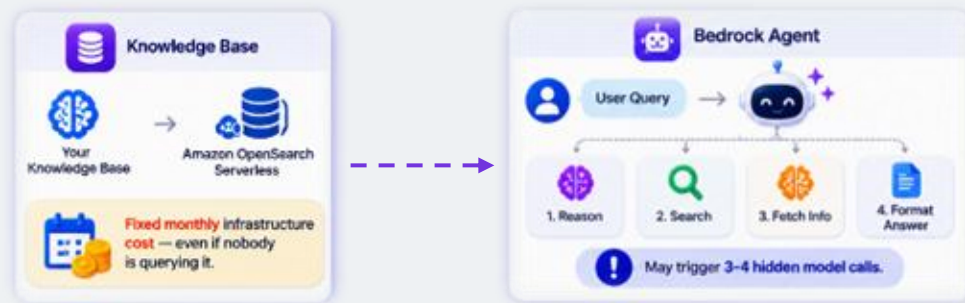
Controlling how much the model writes



Output tokens are almost
always more expensive than
input tokens

Output Optimization & Hidden Infrastructure

Watch for hidden infrastructure costs

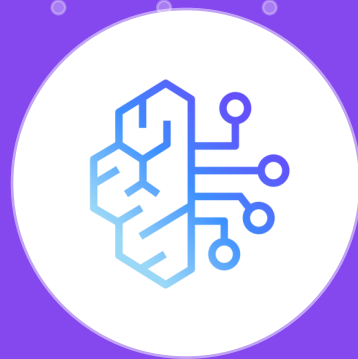


Model calls as the agent
reasons, searches, and formats
its final answer

SECTION INTRODUCTION

Capstone Project: Build HR Document Q & A System

Yogesh Raheja · Thinknyx Technologies LLP



Amazon Bedrock

Section 8 — Capstone Project: Build HR Document Q & A System

Capstone Project: Build HR Document Q & A System

8.0

Section Introduction

8.1

Architecture & Design

8.2

Document Processing

8.3

Vector Database Integration

8.4

Building the Agent

8.5

Enterprise Deployment with AgentCore

Capstone Project: Build HR Document Q & A System

8.0

Section Introduction

8.6

Testing & Best Practices

Problem Statement & Solution Architecture Overview

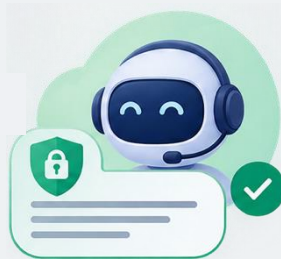


Section 8 › Capstone Project

Problem Statement & Solution Architecture Overview



Problem Statement & Solution Architecture Overview

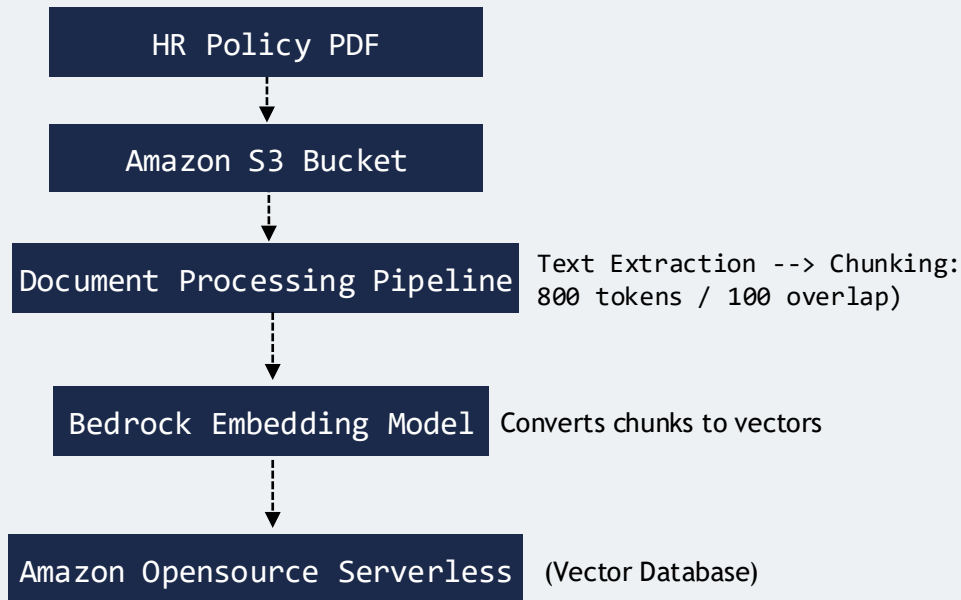


**Private HR Document
Q&A System**

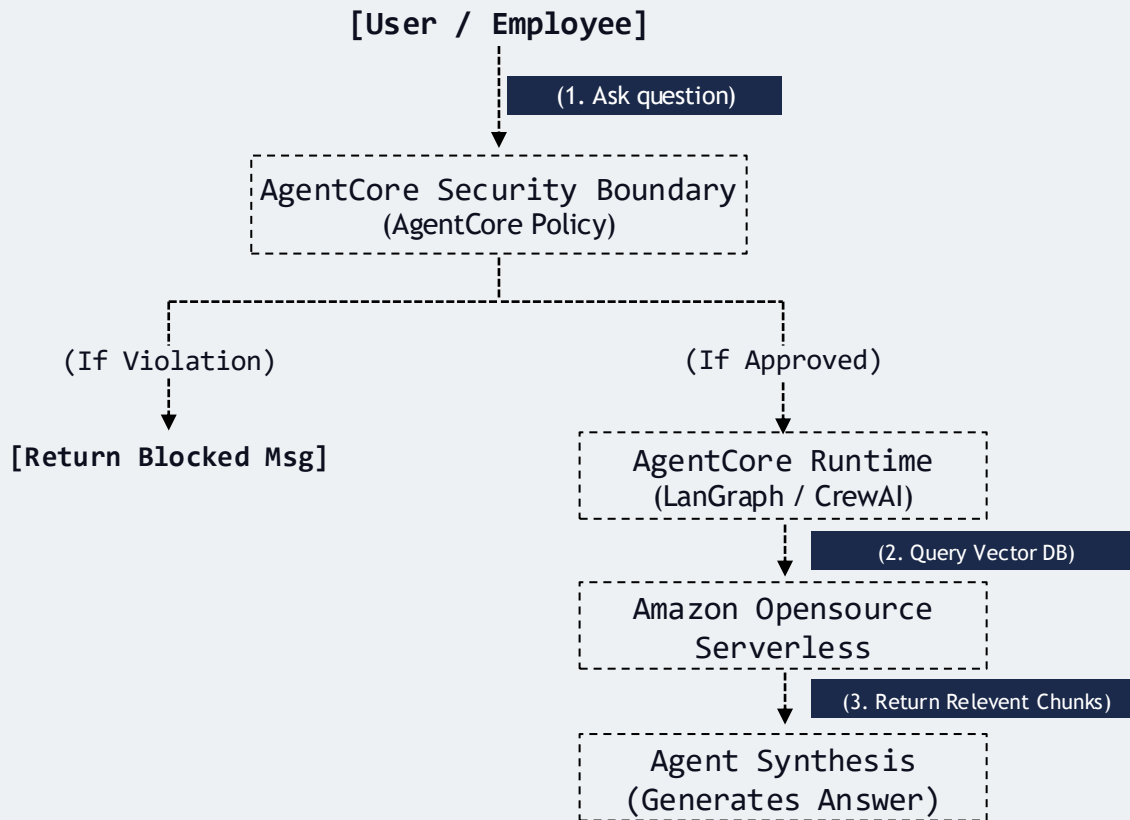
Retrieval-Augmented Generation (RAG) Architecture

Will feed our AI a detailed, real-world HR compliance manual so it can act as a 24/7 intelligent HR assistant

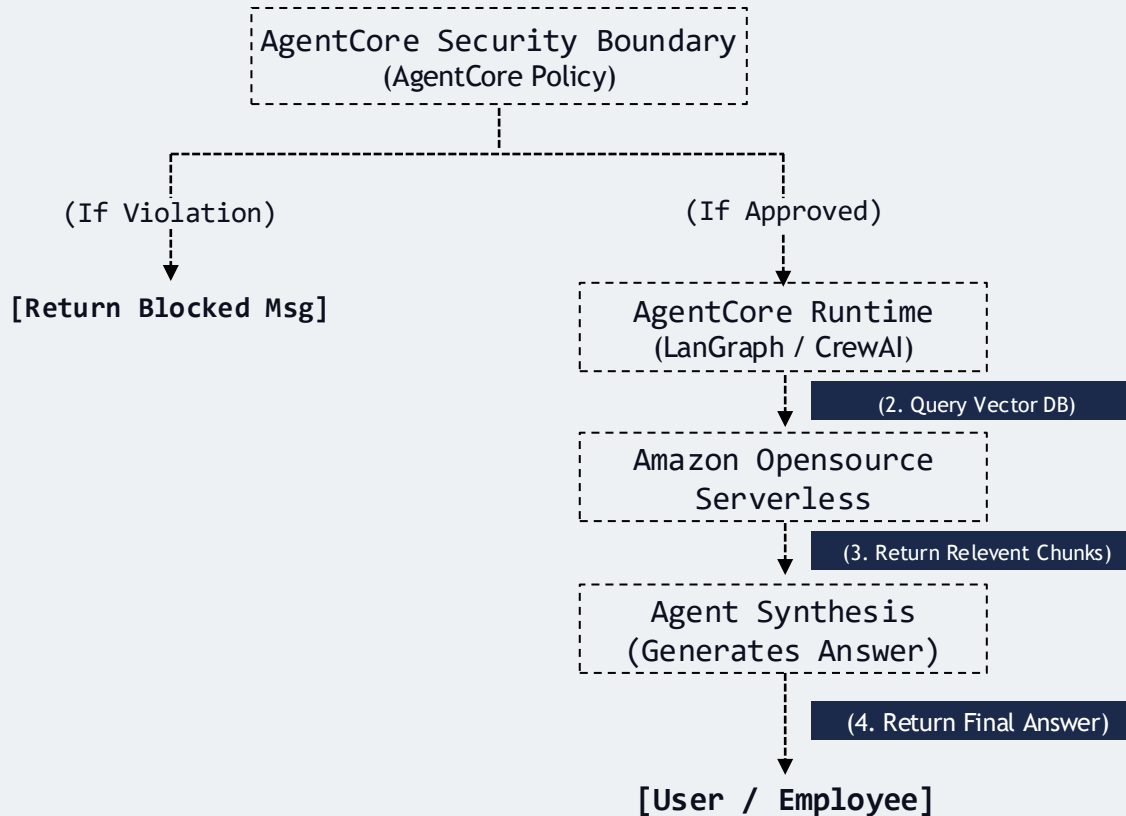
Architecture Overview



Architecture Overview



Architecture Overview

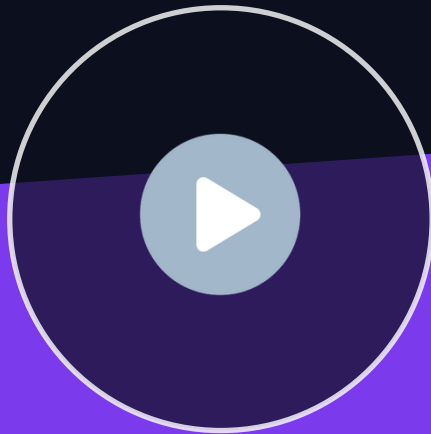


Architecture Overview



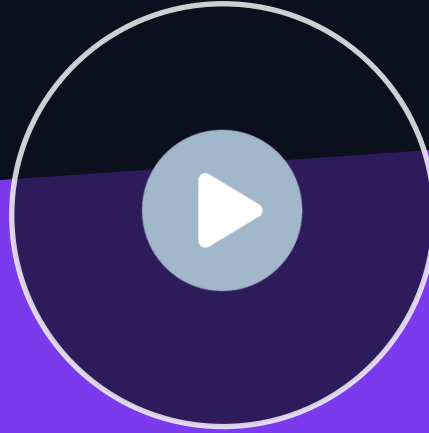
Scalable, production-ready AI agent that seamlessly bridges the gap between raw company data and instant employee support

L I V E D E M O



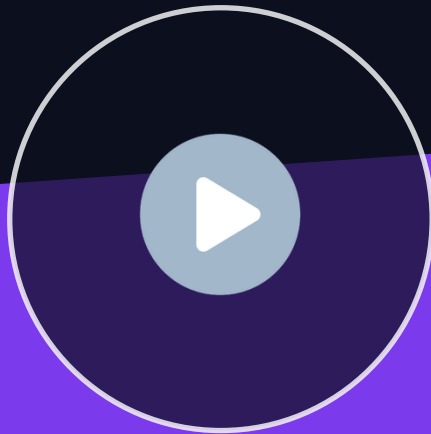
Fine Tuning / Model Customization

L I V E D E M O



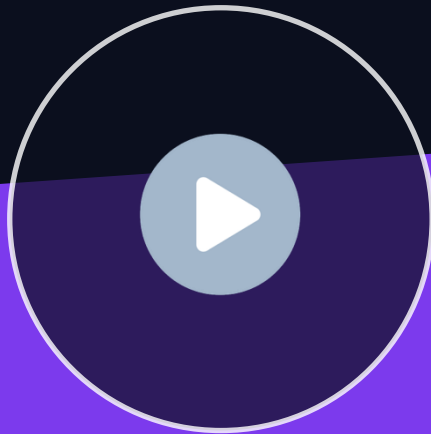
Integration with Vector database

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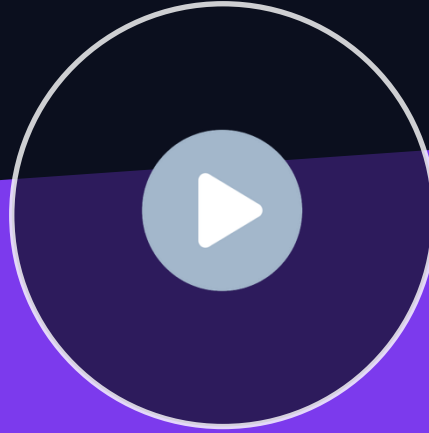
Build a HR document Q&A system

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Deploying to Production with AgentCore

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Testing, Validation, & Observability

Best Practices for RAG Applications



Section 8 › Capstone Project



Secure HR
Q&A agent

Successfully built

Deployed

Tested

Best Practices for RAG Applications

Mastering Your Chunking Strategy

- Answer quality depends on the quality of retrieved text
- 800-token chunks retain deep context without overwhelming the model
- A specific chunk size is the sweet spot, not a random guess
- 100-token overlap prevents concepts from being orphaned across breaks
- Incomplete or confused answers mean you should check chunking first

Best Practices for RAG Applications

Enforcing Hard Security Boundaries

- Data leaks are catastrophic in enterprise environments
- Never rely solely on the AI model's native alignment
- Always implement a deterministic security layer before the LLM
- AgentCore Policies intercept restricted queries before database access
- Treat RAG like an API: authenticate, authorize, and filter

Best Practices for RAG Applications

Optimizing the Context Window

- Never dump retrieved chunks as raw text
- Use strict system prompts to frame the context
- Tell the model exactly how to behave
- Answer strictly using the provided documents
- State when the answer is not in the text to reduce hallucinations

Best Practices for RAG Applications

Continuous Evaluation

- You cannot improve what you do not measure
- Continuously test your RAG system
- Create a golden dataset of common employee questions and verified answers
- Retest whenever you update embeddings, chunk size, or the language model
- Ensure retrieval accuracy does not degrade over time

Build & Secure GenAI Applications with AWS Bedrock: Bootcamp

Yogesh Raheja · Thinknyx Technologies LLP

Amazon Bedrock Course

Conclusion

API Mastery

- Python integration
- Boto3 SDK integration

Prompt Engineering

- Structured output patterns
- Production-ready results

Visual Orchestration

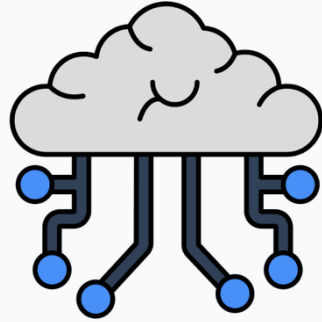
- Automated AI pipelines
- Multi-step Bedrock Flows

Enterprise Governance

- Secure agent infrastructure
- Scale with AgentCore & Guardrails

Production RAG

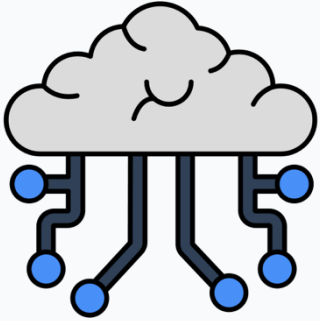
- HR Q&A system architecture
- OpenSearch document pipelines



Generative AI
landscape moves fast



AWS introduce new foundation
models and features



Generative AI
landscape moves fast

Semantic retrieval

Chunking mechanics

Agent orchestration

Safety guardrails



AWS introduce new
foundation models and
features

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